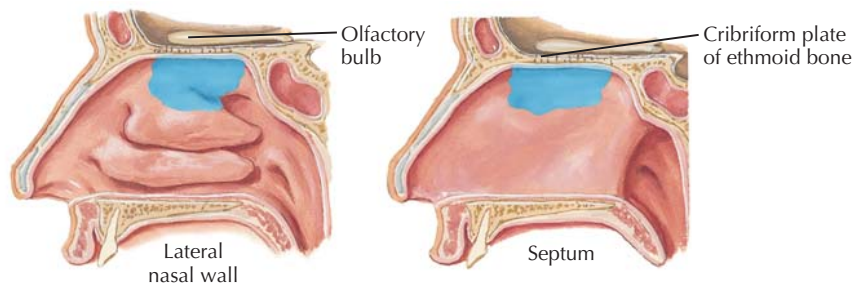
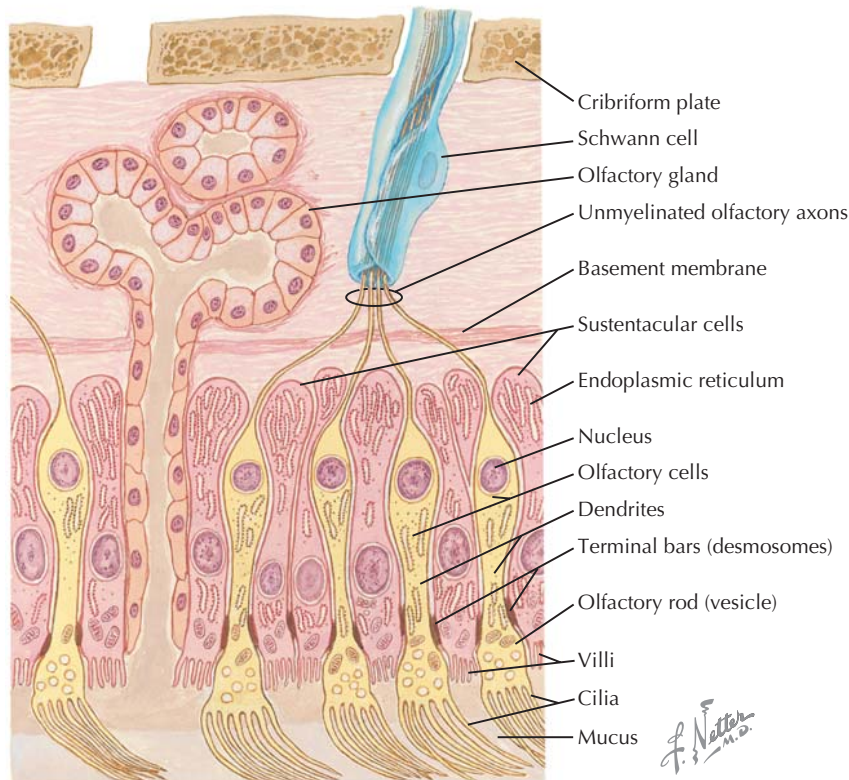


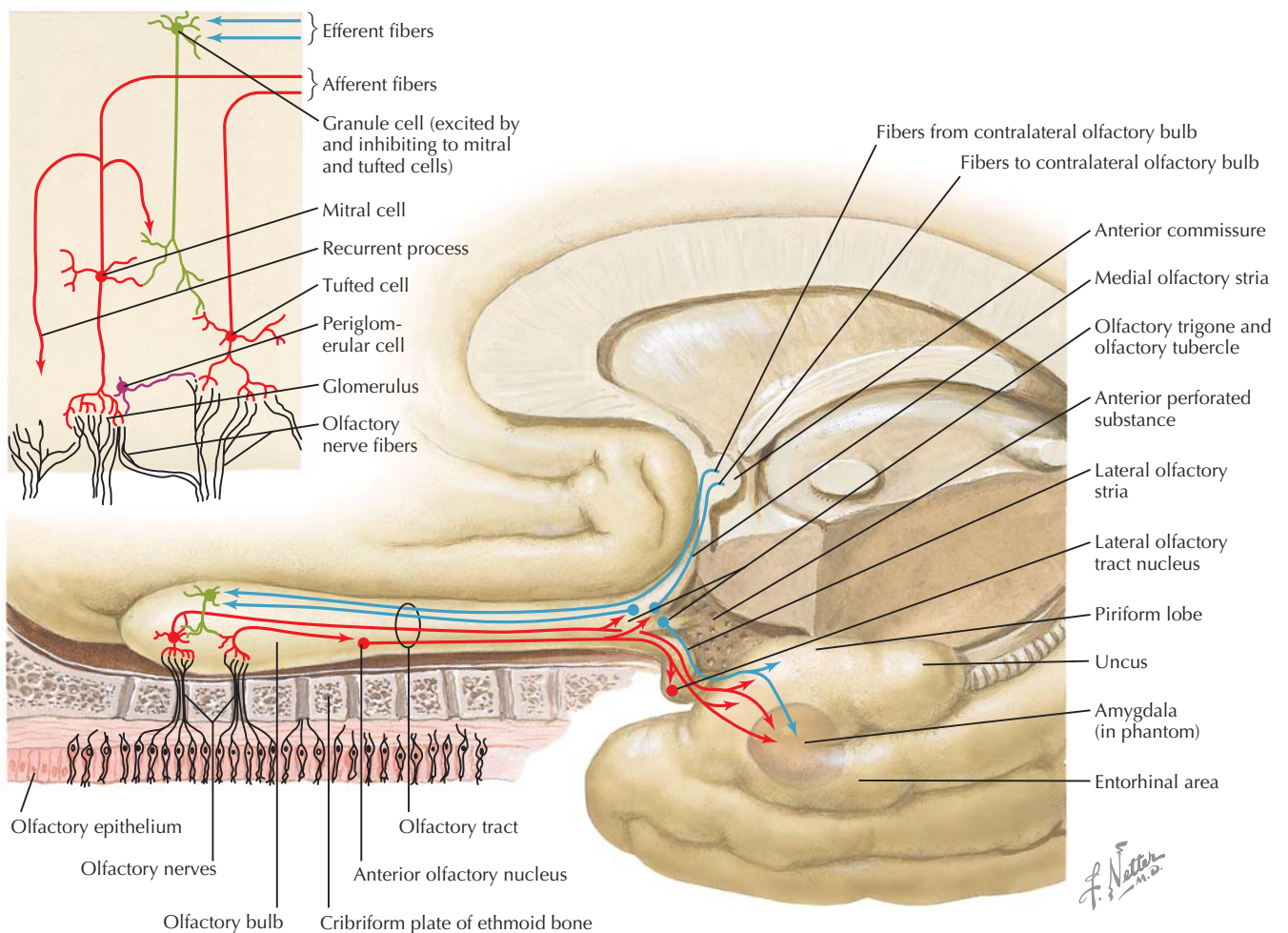
**A. Distribution of olfactory epithelium (blue area)****B. Schema of section through olfactory mucosa**

**Figure 5.14 Olfactory Receptors** Chemical stimuli in the form of odorants result in depolarization of sensory olfactory cells of the olfactory epithelium, located in the upper nasal cavity (A). An olfactory cell, through G protein-coupled receptors, is capable of responding to more than one odor type. Odorants are trapped and also removed by secretions of olfactory glands (B).

## Olfactory Cells

The sense of smell has an important role in appetite in humans, but this sense is poorly developed compared with that of other mammals, in which olfaction can play a major role in reproductive and territorial behavior. The six basic types of odors we sense are camphor, floral, ethereal, musky, putrid, and pungent. The sensory **olfactory cells** are found in the **olfactory epithelium** of the nasal cavity, along with supporting (sustentacular) and basal cells. As is the case with basal cells in taste buds, these basal cells differentiate to continuously replace sensory cells. The cilia of olfactory cells

extend into the nasal mucosa, where they bind odorant molecules on specific G protein-coupled receptors (Fig. 5.14). Binding of an odorant activates signal transduction mechanisms within the olfactory cells, resulting in opening of ion channels and depolarization. Olfactory cells are primary afferent neurons, and action potentials are carried by their axons to the **olfactory bulb**, where they synapse with **mitral cells** (Fig. 5.15). Projections from the mitral cells form the olfactory tract, which carries signals to higher centers, such as the piriform and orbitofrontal cortices, as well as the amygdala.



**Figure 5.15 Olfactory Pathway** Axons of bipolar olfactory nerves synapse with tufted and mitral cell dendrites in glomeruli of the olfactory bulb (detail, including modulatory interneurons, in *inset diagram*). Mitral cell axons form the olfactory tract, which extends to the anterior commissure (where fibers project back to the contralateral olfactory bulb), and the olfactory trigone, ultimately projecting to the primary olfactory cortex, lateral entorhinal cortex, and amygdala.

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# CHAPTER 6

## The Somatic Motor System

The motor system consists of two subdivisions, the **somatic motor system** and the **autonomic motor system**. The somatic nervous system controls **voluntary muscle activity** through motor neurons and skeletal muscle. The autonomic nervous system is responsible for the **involuntary muscle activity** involved in visceral organ function and maintenance of the internal environment.

### MUSCLE SPINDLES

The somatic motor system, through the actions of skeletal muscle, motor neurons, the spinal cord, and the brain, is responsible for the complex task of controlling movement and posture. This task is accomplished through involuntary spinal reflexes and voluntary muscle activity, and is dependent on coordinated contraction and relaxation of muscle. Most skeletal muscle fibers are the **extrafusal fibers** that are innervated by  $\alpha$ -motor neurons and contract to generate force necessary for movement and postural adjustment.

**Intrafusal fibers** are a second category of muscle fibers found specifically within **muscle spindles**; these spindles are distributed in parallel with extrafusal fibers and function as sensory organs that detect length changes in muscle (Fig. 6.1). The sensitivity of intrafusal fibers is regulated by the  $\gamma$ -motor neurons that innervate them. Muscle spindles are abundant in muscles that control fine movements, for example in the eye. They have important roles in coordinating fine movements, in the return of muscles to their normal resting length, and in proprioception. Intrafusal fibers are further classified as **nuclear bag fibers** and **nuclear chain fibers**. Two types of sensory (afferent) nerves innervate intrafusal fibers:

- **Group Ia** afferent nerves innervate both nuclear bag fibers and nuclear chain fibers and transmit information regarding the rate of change in muscle length (contraction). Group Ia fibers form spiral endings around intrafusal fibers.
- **Group II** afferent nerves mainly innervate nuclear chain fibers and detect the length of muscle fibers; they form spraylike endings on the fibers.

**Golgi tendon organs**, like muscle spindles, are stretch receptors and are found at the insertion of skeletal muscles to

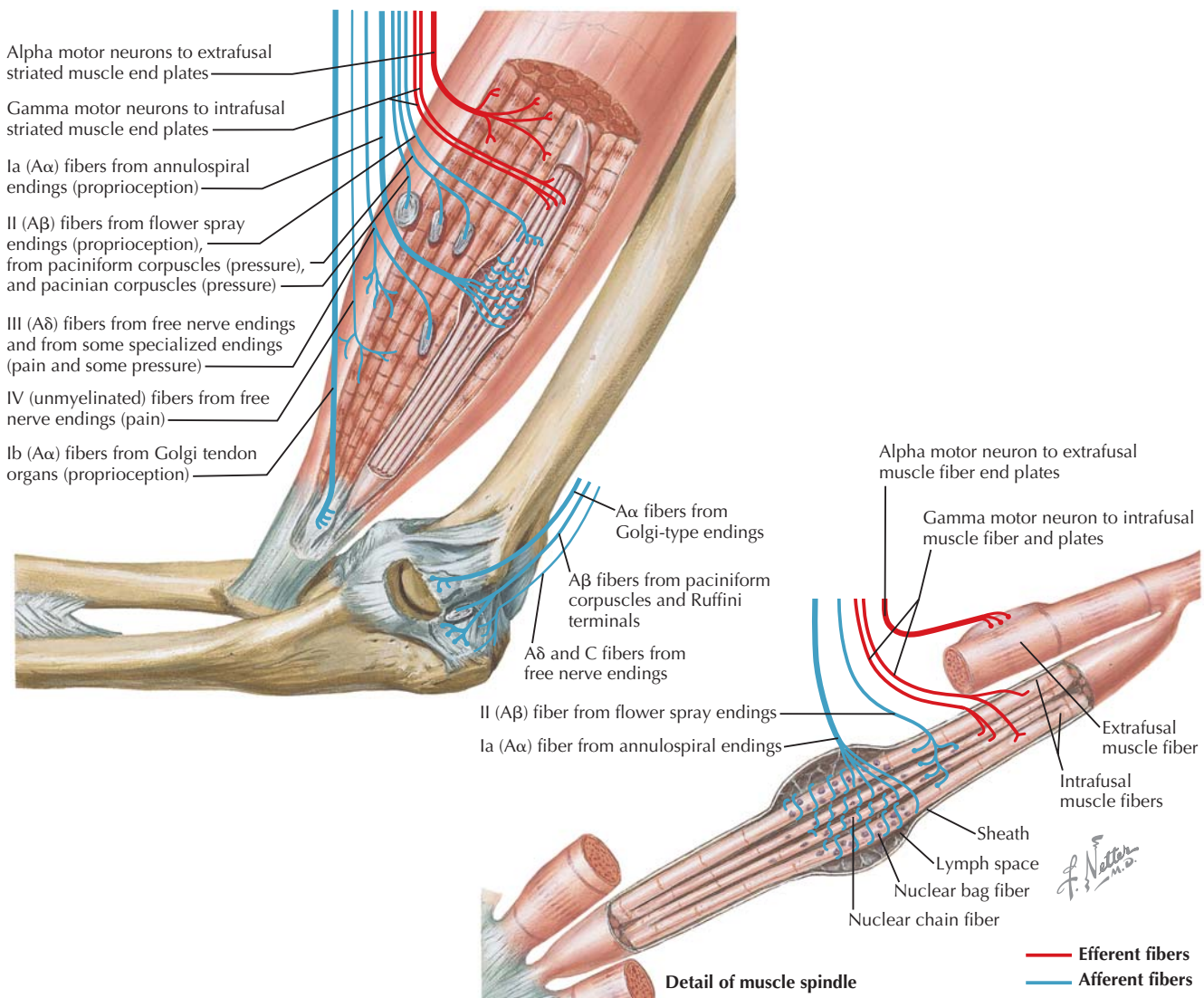
tendons. They are innervated by group Ib afferent nerves. Excessive stretch or contraction of muscle will cause activation of Golgi tendon organs and their afferent nerves, producing a reflexive relaxation of the muscle. Golgi tendon organs are also important in proprioception; their role in this process, along with the role of muscle spindles and other receptors, is illustrated in Figures 6.2 and 6.3.

### SPINAL REFLEXES

Stereotypic motor responses to sensory signals integrated completely within the spinal cord are called **spinal reflexes**. These are the simplest motor responses, but they may be modified by the central nervous system (CNS) during more complex motor activity. The **reflex arc** of a spinal reflex begins with stimulation of sensory receptors; the signal is carried to the spinal cord through sensory afferent nerves, is integrated within the spinal cord, is conducted to the muscle through motor neurons, and finally produces contraction or relaxation. There are three basic types of spinal cord reflexes (Fig. 6.4):

- The **stretch reflex** is the most basic, in which sensory afferent nerves synapse directly with  $\alpha$ -motor neurons in the spinal cord. Stretch reflexes are therefore referred to as **monosynaptic reflexes**. In the **knee jerk reflex**, a type of stretch reflex, a tap on the patellar tendon of the knee stretches muscle spindles within the quadriceps muscle. This initiates the reflex arc, in which type Ia afferent nerves depolarize and conduct the signal to the spinal cord, where they synapse directly with  $\alpha$ -motor neurons. The  $\alpha$ -motor neurons conduct the signal back to the quadriceps, producing contraction (the “knee jerk” response). Simultaneously, activation of interneurons results in relaxation of opposing muscles (see Fig. 6.4A).
- The **Golgi tendon reflex** (also known as the **inverse myotactic reflex**), is a **bisynaptic reflex**. It is a mechanism for prevention of muscle damage when muscle tension is excessive. Stretch of Golgi tendon organs activates type Ib afferent sensory nerves that synapse in the spinal cord with interneurons that subsequently inhibit  $\alpha$ -motor neurons to the muscle, causing relaxation. The reflex also causes antagonistic muscles to contract, coordinating the response (see Fig. 6.4B).



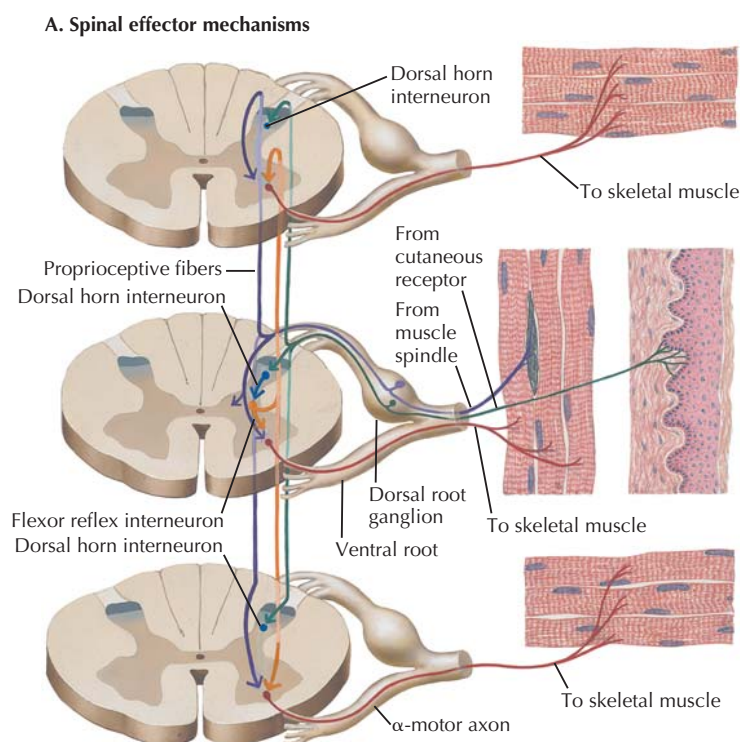


**Figure 6.1 Muscle and Joint Receptors** Muscle spindles are specialized sensory receptors composed of intrafusal fibers of two types, the nuclear chain fiber and the nuclear bag fiber. They function in coordinating fine movements, returning muscles to normal resting length, and proprioception. Intrafusal fibers are innervated by  $\gamma$ -motor neurons that regulate their sensitivity. Information about rate of change in muscle length is conveyed by group Ia afferent sensory nerves innervating both nuclear chain fibers and nuclear bag fibers. These group Ia nerves form spiral endings around intrafusal fibers. Group II sensory afferent nerves convey information on the length of muscle. Golgi tendon organs are located at the insertion of skeletal muscles of tendons; when stretched, they produce a reflexive relaxation of the skeletal muscle. A number of other joint receptors are also illustrated.

- The **Flexor withdrawal reflex** occurs in response to painful or otherwise noxious stimuli. When cutaneous pain receptors are stimulated, for example by touching a hot object, afferent signals are conducted through sensory nerve fibers to the spinal cord, where activation of multiple interneurons produces simultaneous flexion and relaxation of the appropriate muscles to withdraw the limb. This is a **polysynaptic reflex** (see Fig. 6.4C).

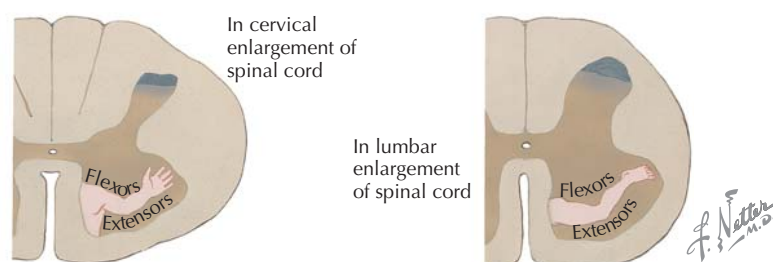
## ROLE OF HIGHER CENTERS IN MOTOR CONTROL

CNS areas involved in motor system control of balance, posture, and movement include the spinal cord, brainstem, cerebellum, basal ganglia, and cerebral motor cortex. For many types of voluntary movement, lower centers regulate patterns of skeletal muscle activity when they are activated by the motor cortex. In the case of fine motor movements, par-



**Figure 6.2 Proprioception: Spinal Effector Mechanisms** A, Spinal effector mechanisms for proprioception involve afferent input from cutaneous receptors and muscle spindles. Monosynaptic reflex pathways at one spinal cord segment and polysynaptic reflex pathways at multiple spinal cord segments activate  $\alpha$ -motor neurons, producing muscle contraction. B, Motor neuron cell bodies producing flexion and extension of limb muscles are distributed somatotopically in the ventral horn.

**B. Schematic representation of motor neurons**



ticularly of the distal muscles of the limbs (for example, in the fingers and hands), the cerebral cortex regulates muscle activity more directly.

### Corticospinal Tract

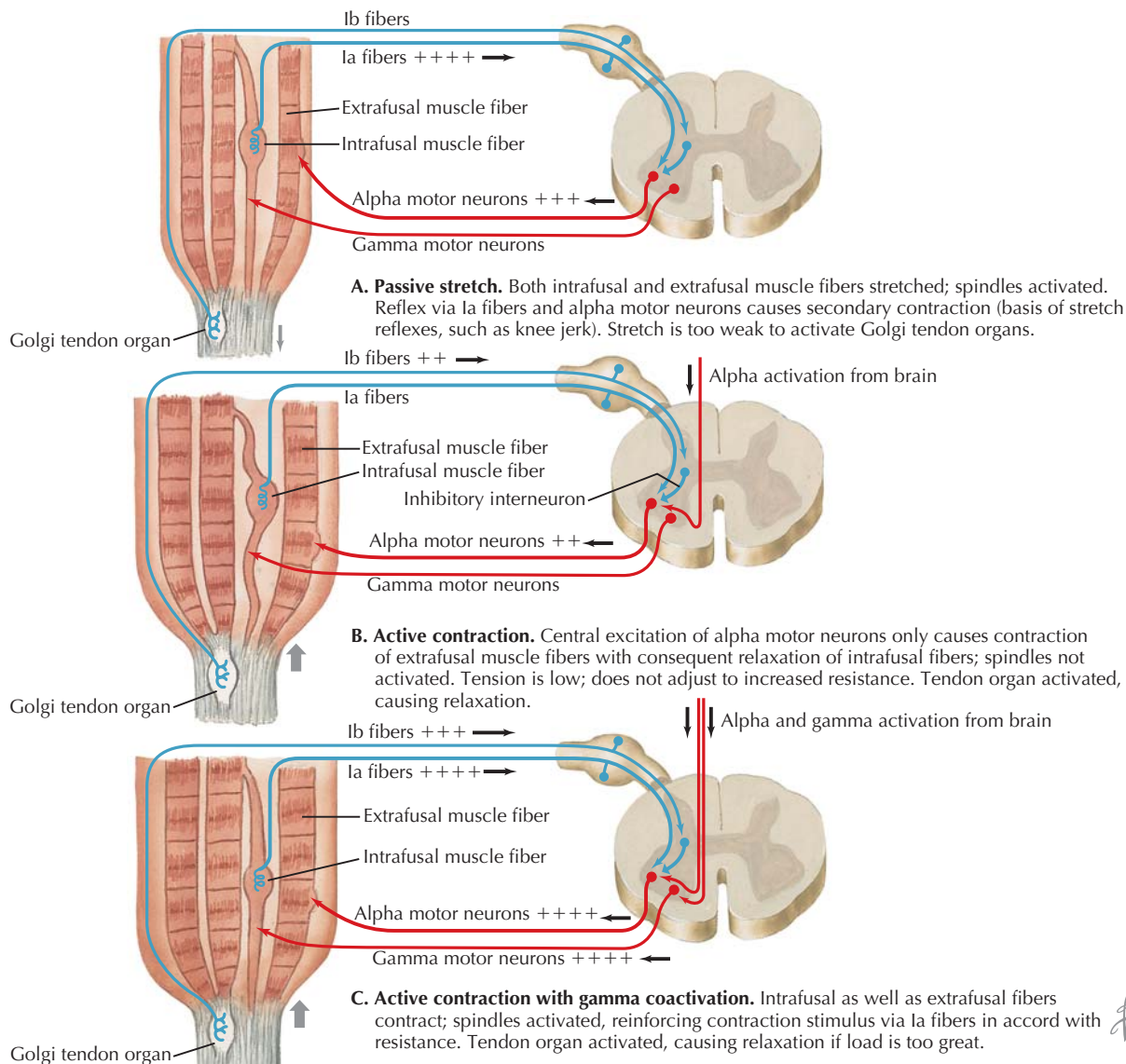
The **corticospinal tract**, also known as the **pyramidal tract** (because it passes through the medullary pyramids), is the most important descending pathway for control of fine motor activity originating in the cortex (Fig. 6.5). Nerve fibers in this tract originate in the **primary motor cortex**, as well as in the adjacent **premotor** and **supplementary motor areas** and somatosensory areas posterior to the motor cortex. Most of the fibers cross in the lower medulla to form the **lateral corticospinal tract**; others descend through the **anterior corticospinal tract**. At various levels of the spinal cord, some fibers from these tracts synapse directly with second order motor

neurons (anterior horn cells). Secondary motor neurons innervated by axons of the lateral corticospinal tract are mainly those controlling distal limb muscles, whereas those innervated by the anterior corticospinal tract are mainly those controlling axial muscles. The **motor homunculus** is a graphic representation of the location and approximate relative size of areas of the motor cortex controlling motor activity of various regions of the body (see Fig. 6.5).

### Brainstem

The brainstem (medulla, pons, and mesencephalon) has an important role in the motor system, especially in the control of balance and posture:

- **Pontine reticular nuclei** send projections through the **pontine reticulospinal tract** of the anterior column of



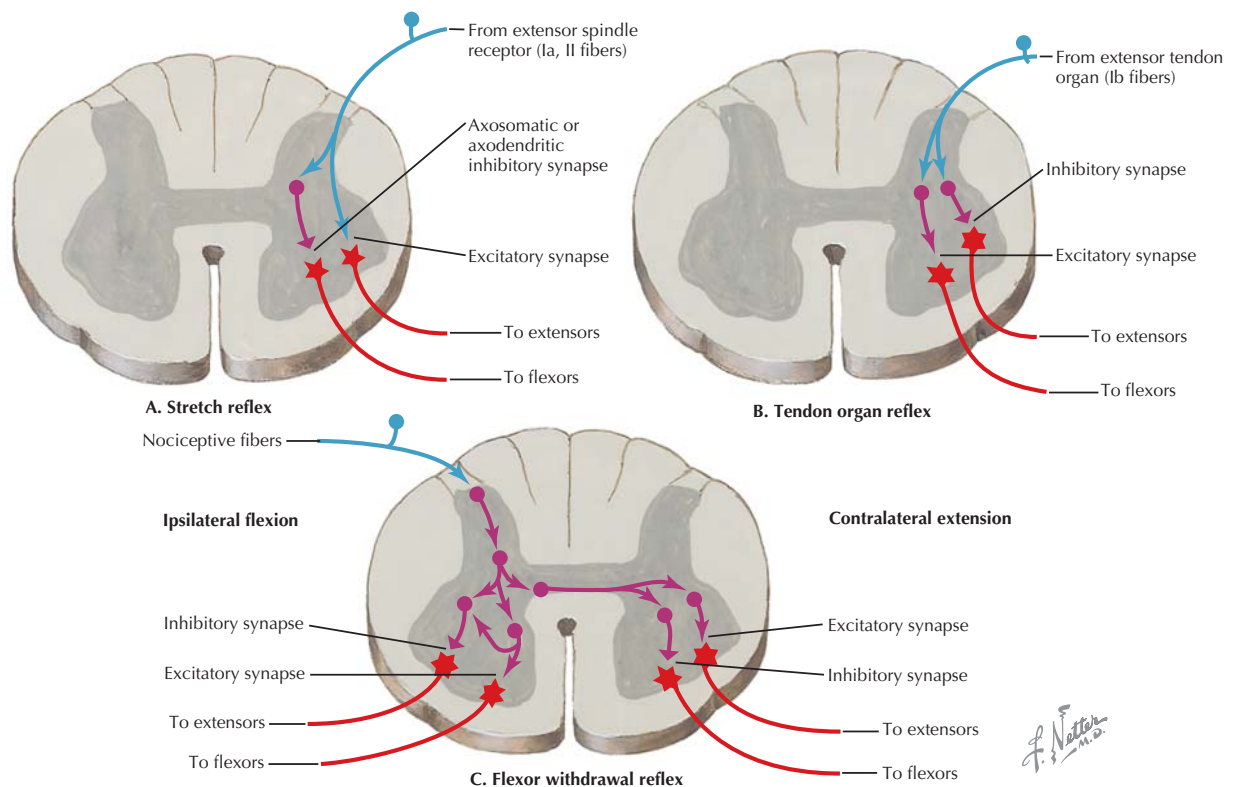
**Figure 6.3** **Proprioceptive Reflex Control of Muscle Tension** A, During passive stretch of skeletal muscle, both intrafusal (as well as extrafusal) fibers are stretched, producing reflex contraction via Ia fibers and  $\alpha$ -motor neurons, for example, during a knee jerk reflex. B and C, Reflex control of skeletal muscle tension in response to active contraction varies depending on state of activation of  $\gamma$ -motor neurons (and its effects on muscle spindles) and degree of stretch of Golgi tendon organs.

the spinal cord, with excitatory synapses on motor neurons innervating axial muscles involved in posture and support of the body against gravity.

- **Medullary reticular nuclei**, through projections descending in the **medullary reticulospinal tract** of the lateral column of the spinal cord, inhibit motor neurons that innervate axial muscles.
- **Vestibular nuclei** on the floor of the fourth ventricle have projections that descend by way of the anterior columns of the spinal cord in the **lateral and medial vestibulospinal tracts**. These nuclei receive input from

the vestibular apparatus and have an important role in controlling muscles to maintain equilibrium, working in conjunction with the pontine reticular nuclei to regulate axial muscles.

- The **superior colliculus** of the mesencephalon receives input from visual nuclei; through the **tectospinal tract**, it is involved in control of head, neck, and eye muscles.
- The **red nucleus** of the mesencephalon receives input from the motor cortex. It sends projections in the **rubrospinal tract** through the lateral columns of the spinal



**Figure 6.4 Spinal Reflex Pathways for Stretch, Tendon Organ, and Flexor Withdrawal Reflexes** Reflexes elicited by stretch of muscle spindle receptor (A) or Golgi tendon organs (B) associated with an extensor muscle are illustrated. Opposite effects on flexors and extensors occur for reflexes initiated by stretch of a flexor muscle or tendon organ. In the flexor withdrawal reflex, nociceptive stimuli produce ipsilateral flexion and contralateral extension through the pathways illustrated (C).

cord and is involved in voluntary control of large muscles, particularly in the limbs. It can be considered in some ways as an alternative pathway to the corticospinal tract for controlling voluntary muscle movement. This is ordinarily a relatively minor pathway in humans, compared with other mammals, but may gradually assume an expanded role when the corticospinal tract is injured.

## Cerebellum

The **cerebellum** (“little brain” in Latin) has an important accessory role to the motor cortex, in coordination and control of posture, balance and movement, and in planning and initiation of motion. These effects are accomplished through the activity of its three lobes (Fig. 6.6):

- The **archicerebellum** (also known as the vestibulocerebellum) is involved in the regulation of posture and balance, and in control of eye and head movement. It receives afferent signals from the vestibular apparatus and sends efferent signals through the relevant descending efferent pathways (Fig. 6.7).

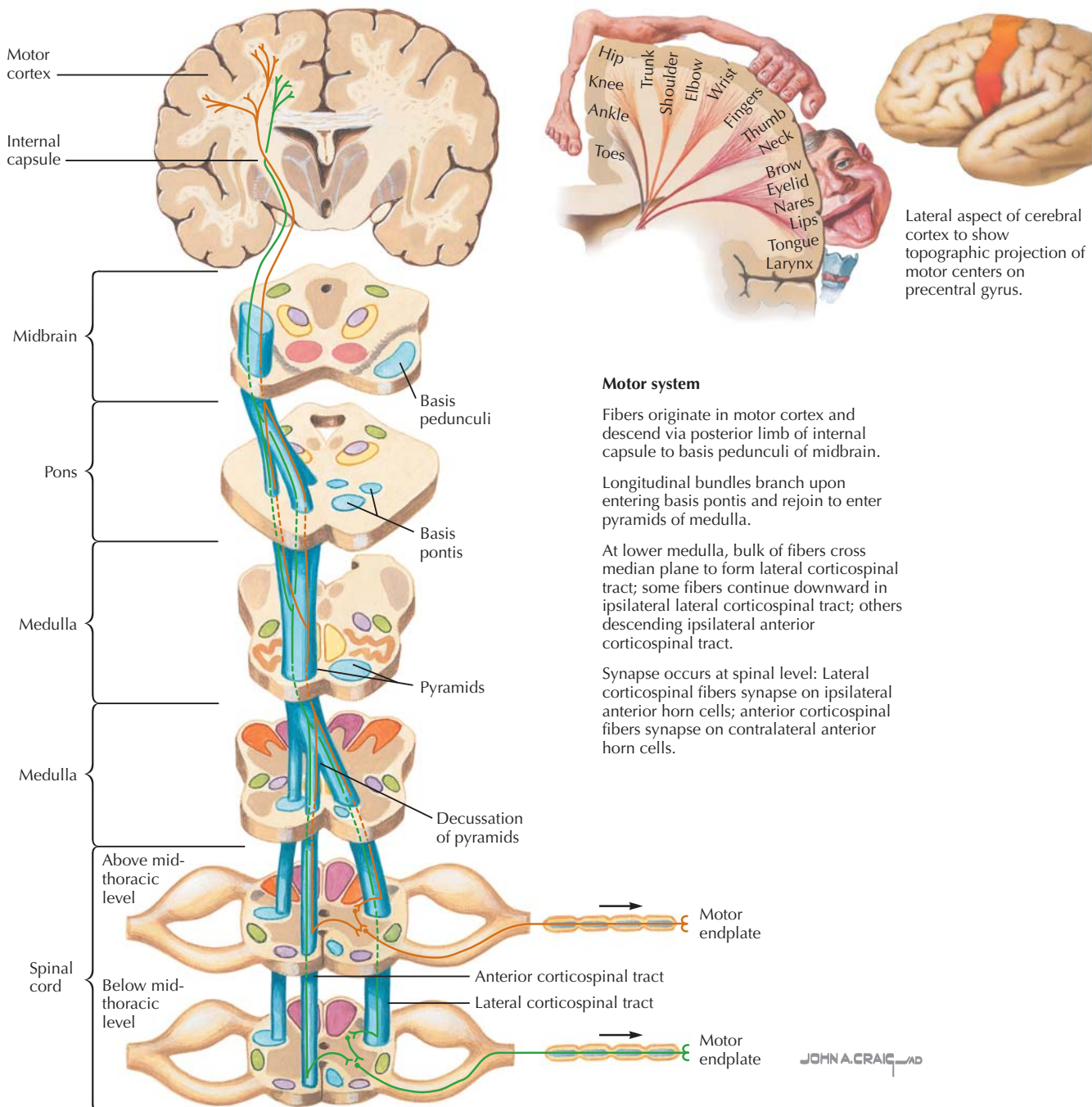
- The **paleocerebellum** (also called the **spinocerebellum**) has an important role in the regulation of proximal limb movement. Afferent sensory signals regarding the position and movement of limbs are utilized to “fine-tune” limb motion through the relevant descending pathways (see Fig. 6.7).
- The **neocerebellum** (also known as the **pontocerebellum**) has a coordinating role in the regulation of distal limb movement, based on input from the cerebral cortex (via the pontine nuclei). It aids in the planning and initiation of motor activity through its efferent fibers (see Fig. 6.7).

Alternatively, the lobes of the cerebellum can be conceptualized as anterior, middle, and flocculonodular lobes (see Fig. 6.6).

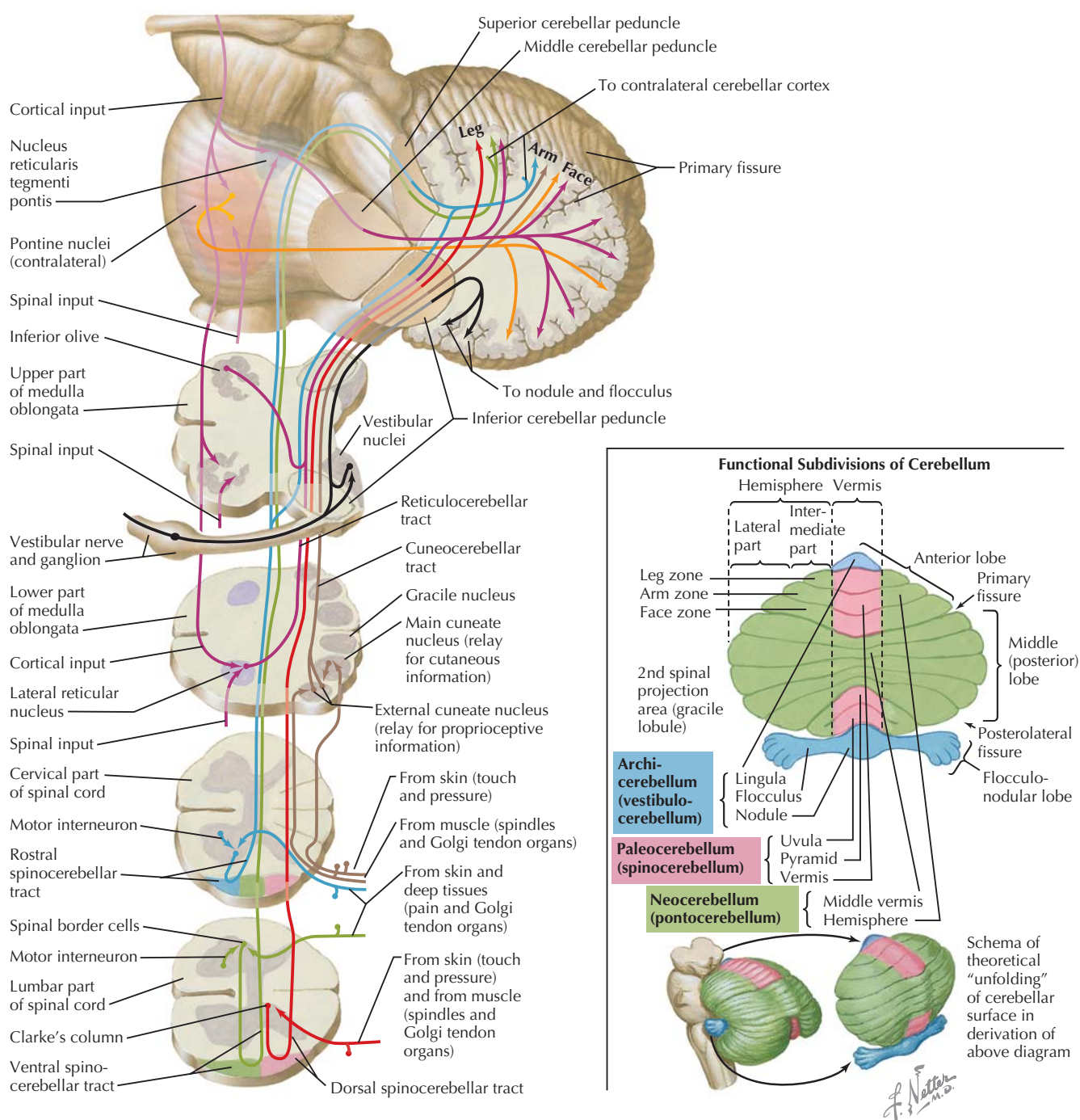


The cerebellum’s role in muscle control involves coordination and fine-tuning of motion. Deficits in cerebellar function are expressed as deficiency in fine motor activity, coordination, and equilibrium.

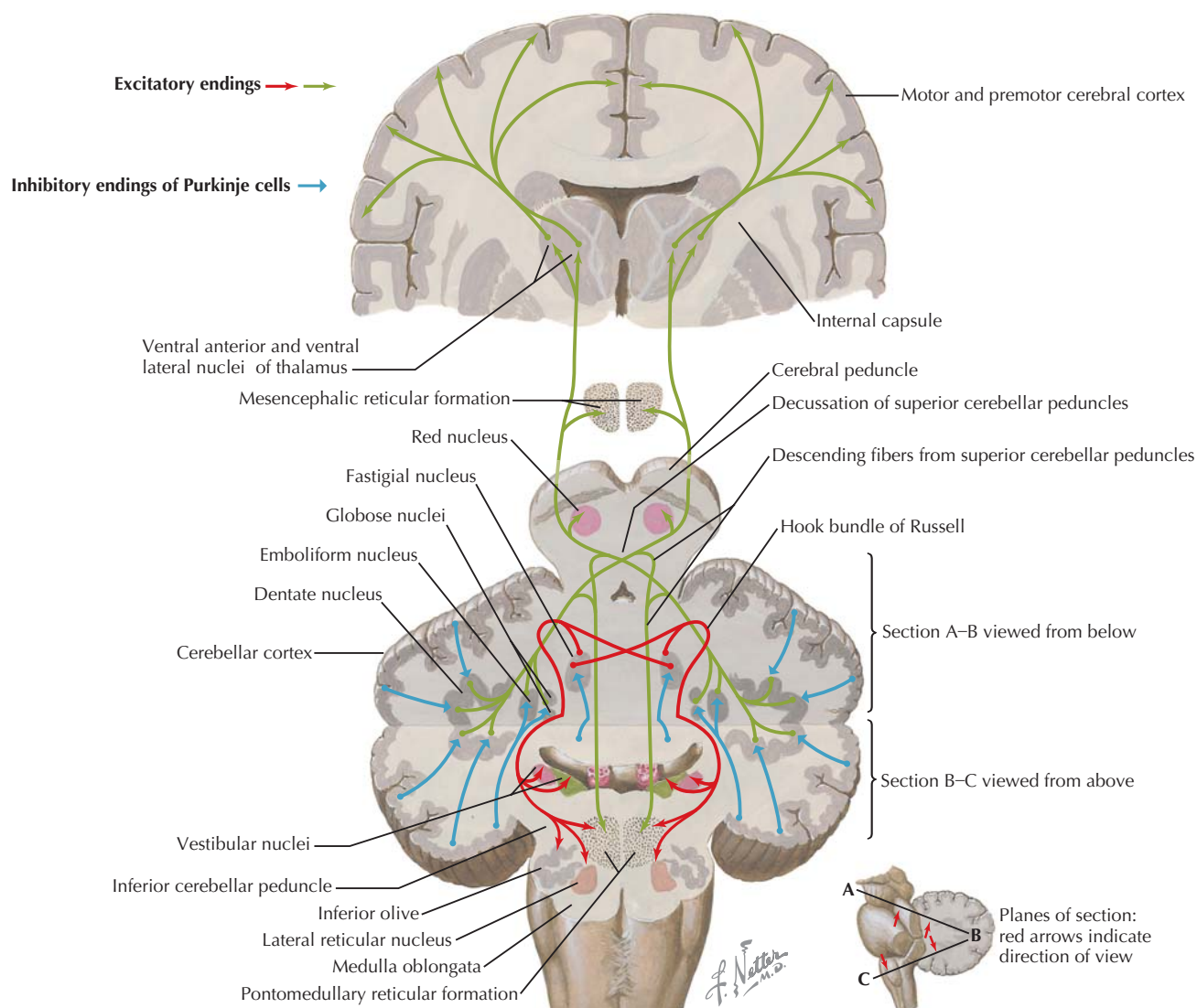




**Figure 6.5 Corticospinal Tract** The corticospinal tract, also called the pyramidal tract, is the major descending pathway in voluntary motor activity and is particularly important in fine motor activity. The motor homunculus (*top center panel*) illustrates the location and approximate relative size of areas in the motor cortex controlling muscle activity in various regions of the body.



**Figure 6.6 Cerebellar Afferent Pathways and Functional Subdivisions of Cerebellum** The cerebellum acts as an accessory to the motor cortex in regulation of posture, balance, movement, and planning and initiation of motion, based on afferent sensory signals and cortical input. Specific functions ascribed to subdivisions of the cerebellum are illustrated in the right panel or discussed in the text.



**Figure 6.7 Cerebellar Efferent Pathways** The cerebellar efferent pathways originate from its deep nuclei, which receive inhibitory input from Purkinje cells of the cerebellar cortex. Axons project from the deep fastigial, globose, emboliform, and dentate nuclei to various nuclei in the brainstem, midbrain, and thalamus, and modulate activity of descending motor pathways, resulting in coordination and fine-tuning of motion.

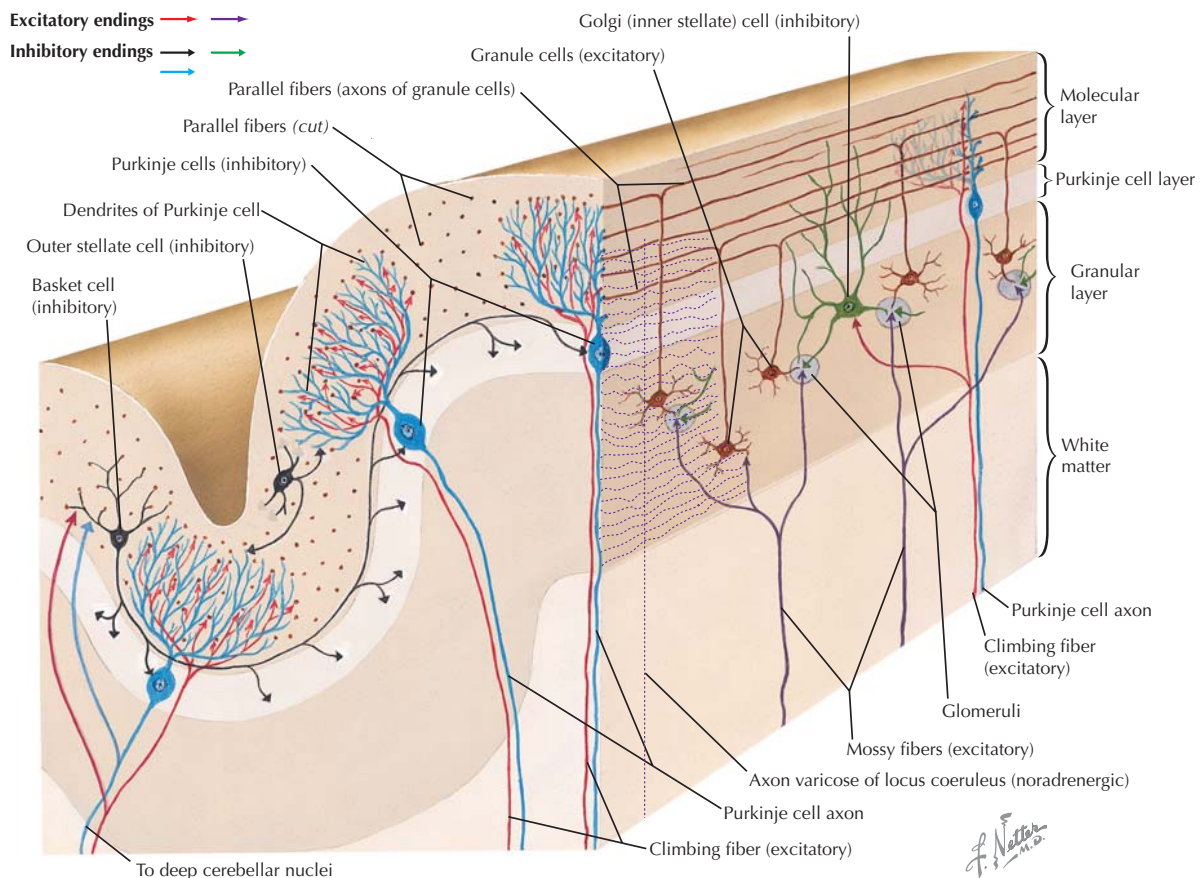
### Cerebellar Cortex

All efferent signals of the cerebellar cortex are inhibitory and emanate from **Purkinje cells** residing in the middle of three layers of the cortex. Signals from Purkinje cells are conducted to deep cerebellar nuclei (see Fig. 6.7). The three layers of the cortex contain various cell types that, through complex interactions between each other and with fibers projecting from other sites, regulate the inhibitory efferent output of Purkinje cells (Fig. 6.8):

- The inner **granular layer**, containing **granule cells**, **Golgi cells**, and **glomeruli**. Granule cells within this

layer are the most numerous type of neuron in the CNS. The glomeruli are the sites of synapses between axons of **mossy fibers** from the spinocerebellar, vestibulocerebellar, and pontocerebellar tracts with the dendrites of the granule and Golgi cells. Granule cells, which receive excitatory input from the mossy fibers, send axons through the Purkinje cell layer to the outer molecular layer (discussed later). They provide excitatory signals to Purkinje cell dendrites and dendrites of cells of the molecular layer. Golgi cells are inhibitory interneurons that inhibit the granule cells' effects on Purkinje cells.





**Figure 6.8 Cerebellar Neuronal Circuitry** The cerebellar cortex is composed of three layers. All output of the cortex is carried by inhibitory axons of the Purkinje cells. Their cell bodies are located in the Purkinje cell layer, and their dendritic arbors are in the outer molecular layer. The cortex receives input through climbing fibers, which project from the medullary inferior olive, and mossy fibers, which project from the spinocerebellar, vestibulocerebellar, and pontocerebellar tracts. Mossy fibers form excitatory synapses with dendrites of granule cells in the granular layer. The excitatory axons of granule cells reach the molecular layer, where they form parallel fibers that course through and synapse with dendritic arbors of many Purkinje cells. Climbing fibers form excitatory synapses directly with dendrites of Purkinje cells in the molecular layer. Various other interneurons and their actions are illustrated above and discussed in the text.

- The middle **Purkinje cell layer**. Purkinje cells are GABA-ergic (utilize the neurotransmitter gamma amino butyric acid) and are the only efferent output of the cerebellar cortex. They receive input from various axons in the molecular layer and send inhibitory signals to the deep cerebellar nuclei and lateral vestibular nuclei.
- The outer **molecular layer**, which contains basket cells and stellate cells, and the dendritic arbors of Purkinje cells (extending from the Purkinje cell layer). **Parallel fibers**, containing axons from the granule cells of the granular cell layer, form excitatory synapses with dendrites of Purkinje, basket, and stellate cells. Basket cells and stellate cells are interneurons that inhibit Purkinje cells through the parallel fibers. **Climbing fibers**, projecting from the medullary inferior olive,

form excitatory synapses with dendrites of Purkinje cells.

Thus, the inhibitory cerebellar cortical output of the Purkinje cells is regulated by excitatory input from climbing fibers and mossy fibers (via the granule cells) and inhibitory input from various interneurons of the cortex.



Purkinje cells in the cerebellar cortex are stimulated by granule cells and climbing fibers; their activity is modulated by various cerebellar interneurons. When stimulated, Purkinje cells release the inhibitory neurotransmitter GABA at synapses within deep cerebellar nuclei, regulating motion and posture.



## Basal Ganglia

The **basal ganglia** are deep nuclei of the cerebral hemispheres, and like the cerebellum, have an accessory role to the motor cortex, regulating motor activity to produce smooth movement and maintain posture. They consist of the:

- caudate nucleus
- putamen
- globus pallidus

The caudate nucleus and putamen are referred to as the **striatum**, based on their striated appearance. The **substantia nigra**

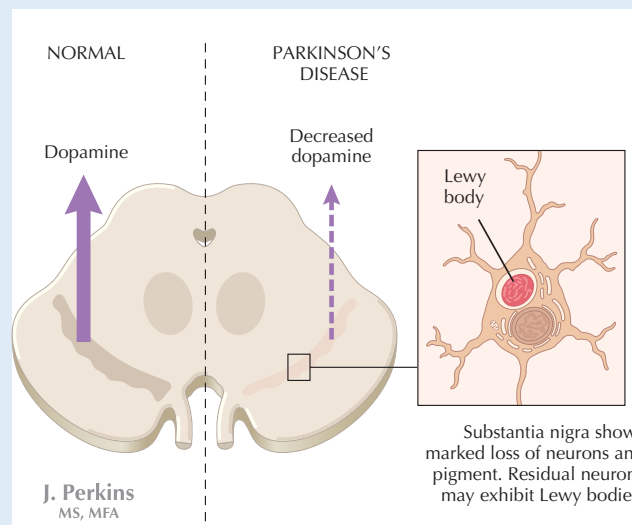
of the midbrain and **subthalamic nucleus** of the diencephalon are functionally associated with, and sometimes classified as, components of the basal ganglia.

The input to the basal ganglia is from the motor cortex; the output to the cortex is through the thalamus. A complex series of interactions between components of the basal ganglia and associated structures leads to this output, which then regulates the level of excitation in the motor cortex. These interactions form an **indirect pathway** that inhibits cortical excitation and a **direct pathway** that is excitatory. The balance of these opposing pathways is responsible for coordinating smooth movement and maintenance of posture.

### CLINICAL CORRELATE Parkinson's Disease

Diseases of the basal ganglia produce a variety of deficits in motor activity and control of posture. Parkinson's disease is a progressive neurodegenerative disorder characterized by tremor of the hands and arms, rigidity, shuffling gait, and bradykinesia (slow move-

ment). This disorder is caused by loss of dopaminergic cells of the substantia nigra normally projecting to the striatum (caudate nucleus and putamen), affecting both the direct and indirect pathways of the basal ganglia. L-dopa, a dopamine precursor, is effective in treating some but not all Parkinsonian patients.



**The Substantia Nigra in Parkinson's Disease** Dopamine production by the substantia nigra is reduced in Parkinson's disease as a result of degeneration of dopaminergic neurons. Lewy bodies are observed in the remaining neurons of the substantia nigra of some Parkinson's disease patients. These inclusions are caused by cellular accumulation of the protein alpha-synuclein.

# CHAPTER 7

## The Autonomic Nervous System

Knowledge of the role of the autonomic nervous system is critical for understanding the function of any of the major organ systems. It is the primary mechanism for the involuntary control and coordinated activity of smooth muscle of visceral organs, cardiac muscle, and glands and is essential for most homeostatic processes. Within the brain, sensory information is integrated and activity of the autonomic nervous system is modulated to orchestrate this involuntary control of physiological processes.

### ORGANIZATION AND GENERAL FUNCTIONS OF THE AUTONOMIC NERVOUS SYSTEM

The two divisions of the autonomic nervous system are the **sympathetic nervous system (SNS)** and the **parasympathetic nervous system (PNS)** (the enteric nervous system of the gastrointestinal tract is sometimes considered part of the autonomic nervous system as well; it is discussed in Chapter 21). In many cases, the SNS and PNS have opposing actions on various organs and processes, and regulation of bodily functions often involves reciprocal actions of the two divisions. For example, heart rate is elevated by SNS activity and decreased by PNS activity.

As a generalization, the SNS is said to mediate **stress responses**, such as the classic *fight-or-flight* response, and the PNS mediates “vegetative” responses, such as digestion. The fight-or-flight response is a generalized reaction to extreme fear, stress, or physical activity and results in a patterned response in many organ systems. The response includes elevated heart rate, cardiac output, and blood pressure, as well as bronchial dilation, mydriasis (dilation of pupils), and sweating. Although the sympathetic nervous system often responds in such patterned manners, the parasympathetic may produce more selective effects, for example during the sexual response.

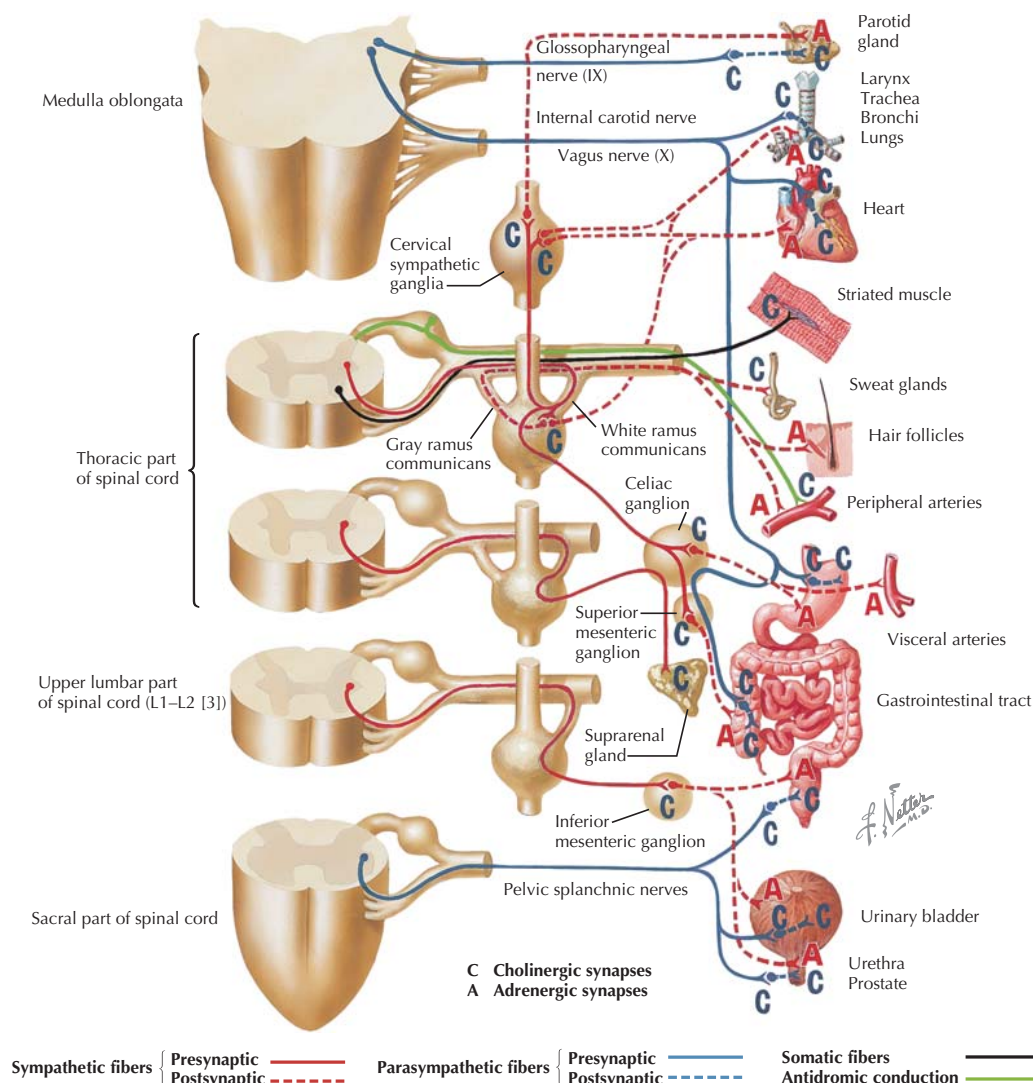
The autonomic nervous system has as its central components the hypothalamus, the brainstem, and the spinal cord; peripherally, it consists of sympathetic and parasympathetic nerves. Areas within the hypothalamus and brainstem regulate and coordinate various processes through the autonomic nervous system, including, for example, temperature regulation, responses to thirst and hunger, micturition, respiration, and cardiovascular function. This regulation is in response to



The fight-or-flight response was originally described in 1915 by Walter Canon, who also invented the term “homeostasis.” The fight-or-flight response can be characterized as the physiological response to acute stress, in which generalized sympathetic activation occurs, resulting in effects such as tachycardia, bronchial dilation, mydriasis (dilation of the pupils), vasoconstriction in much of the body, piloerection, and inhibition of gastrointestinal motility. It has long been appreciated that the acute stress responses also involves activation of the hypothalamic-pituitary-adrenocortical endocrine axis (Section 7).

sensory input and occurs through the reciprocal regulation of the SNS and PNS.

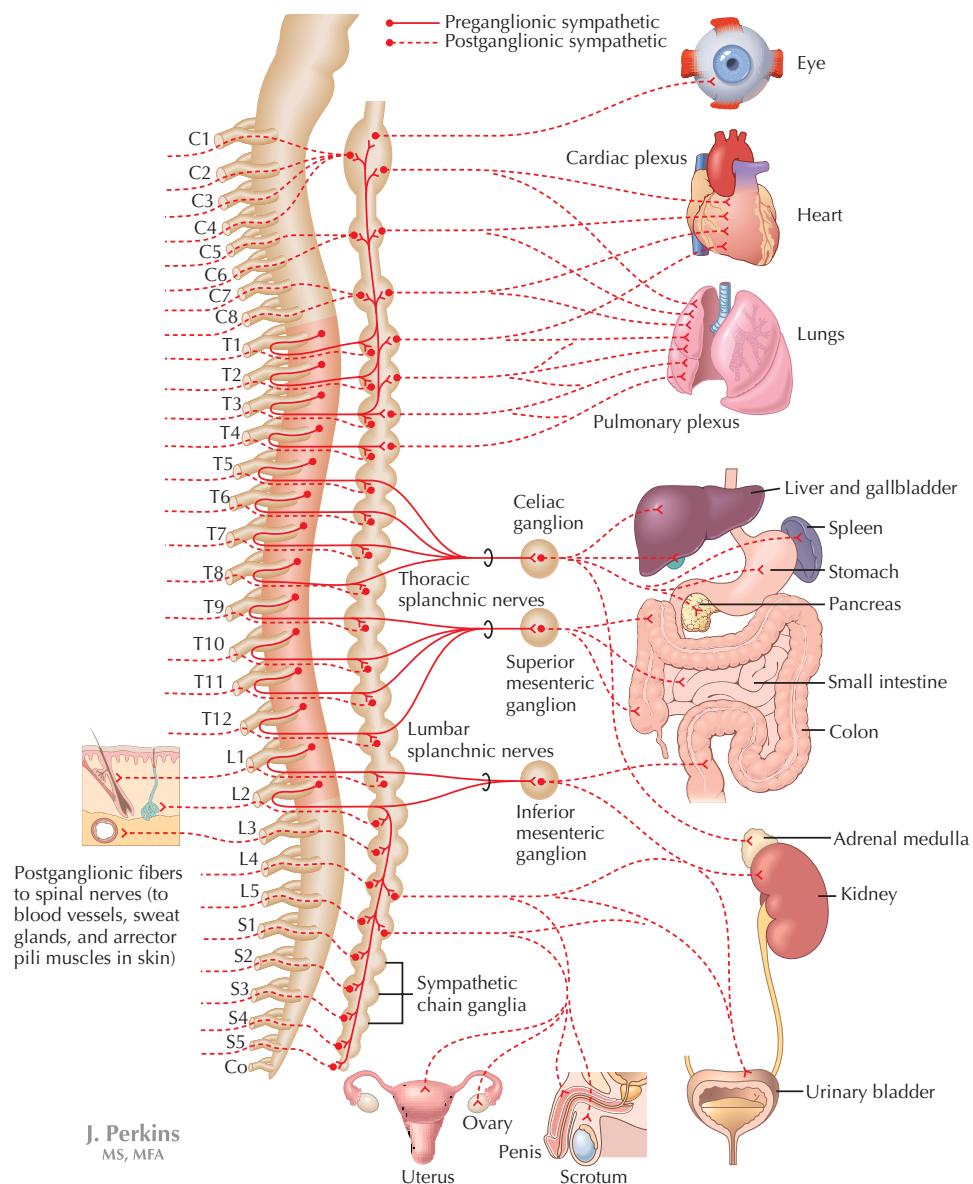
Peripherally, axons of **preganglionic neurons** of the SNS and the PNS emerge from the spinal cord and synapse with **postganglionic neurons** at sympathetic and parasympathetic ganglia, respectively, where in both cases, **acetylcholine** is the neurotransmitter, acting at **nicotinic receptors** on postganglionic neurons (Fig. 7.1). Postganglionic neurons then send motor axons to effector organs and tissues. The catecholamine **norepinephrine** is released by postganglionic sympathetic axons and acts at **adrenergic receptors** of effector organs. One exception is the postganglionic axons innervating sweat glands, which release acetylcholine. Furthermore, the adrenal medulla functions as part of the SNS. Preganglionic axons of the SNS extend to the adrenal gland, where they stimulate chromaffin cells of the adrenal medulla to release **epinephrine** (and to a lesser degree norepinephrine) into the bloodstream. Notably, in addition to the catecholamines (norepinephrine and epinephrine), a number of **adrenergic cotransmitters** are released by some sympathetic postganglionic nerves, including **neuropeptide Y**, **ATP** (adenosine triphosphate), and **substance P**, among others. In the PNS, acetylcholine, acting at **muscarinic receptors**, is the postganglionic neurotransmitter. These and other aspects of the two divisions of the autonomic nervous system are compared in Table 7.1 and illustrated in Figures 7.2 and 7.3. Actions of the autonomic nervous system in various organ systems and tissues are listed in Table 7.2, along with the receptor types involved.



**Figure 7.1 Cholinergic and Adrenergic Synapses: Schema** Preganglionic neurons of the autonomic nervous system emerge from the spinal cord and synapse at autonomic ganglia. For both parasympathetic and sympathetic divisions, acetylcholine is the ganglionic neurotransmitter. Postganglionic fibers from parasympathetic ganglia release acetylcholine, which binds to muscarinic receptors at the effector organs. Sympathetic postganglionic axons release mainly norepinephrine, although acetylcholine is released at sweat glands. The adrenal gland (not illustrated) functions as part of the SNS. Chromaffin cells of the adrenal medulla act as postganglionic cells of the SNS, releasing epinephrine (and to some extent, norepinephrine) directly into the blood stream. Sympathetic postganglionic nerves, in addition to releasing catecholamines (norepinephrine and epinephrine), release several cotransmitters, including neuropeptide Y, ATP, and substance P.

**Table 7.1 General Characteristics of the Parasympathetic and Sympathetic Nervous Systems**

| Characteristic                                           | Parasympathetic Nervous System                                                                                        | Sympathetic Nervous System                                                                   |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Location of preganglionic nerve cell bodies              | Brainstem (nuclei of cranial nerves II, VII, IX, and X) or sacral spinal cord (S2–S4; sacral parasympathetic nucleus) | Intermediolateral and intermediomedial cell columns of the thoracolumbar spinal cord (T1–L3) |
| Location of ganglia                                      | In or adjacent to target organs                                                                                       | Paravertebral and prevertebral                                                               |
| Neurotransmitter of preganglionic neurons                | Acetylcholine (acts at nicotinic receptors)                                                                           | Acetylcholine (acts at nicotinic receptors)                                                  |
| Major neurotransmitter released by postganglionic neuron | Acetylcholine (acts at muscarinic receptors)                                                                          | Norepinephrine (acts at $\alpha$ and $\beta$ adrenergic receptors)                           |



**Figure 7.2 Autonomic Nervous System: Sympathetic Division** The two divisions of the autonomic nervous system are the sympathetic nervous system and the parasympathetic nervous system. Preganglionic fibers of the SNS emerge from the spinal cord at levels T1–T2. The SNS is involved in “fight-or-flight” responses, as well as responses to exercise, hemorrhage, and other challenges to homeostasis. Both SNS and PNS innervate smooth and cardiac muscle and glands; in general, they work together in reciprocal fashion to regulate bodily function.

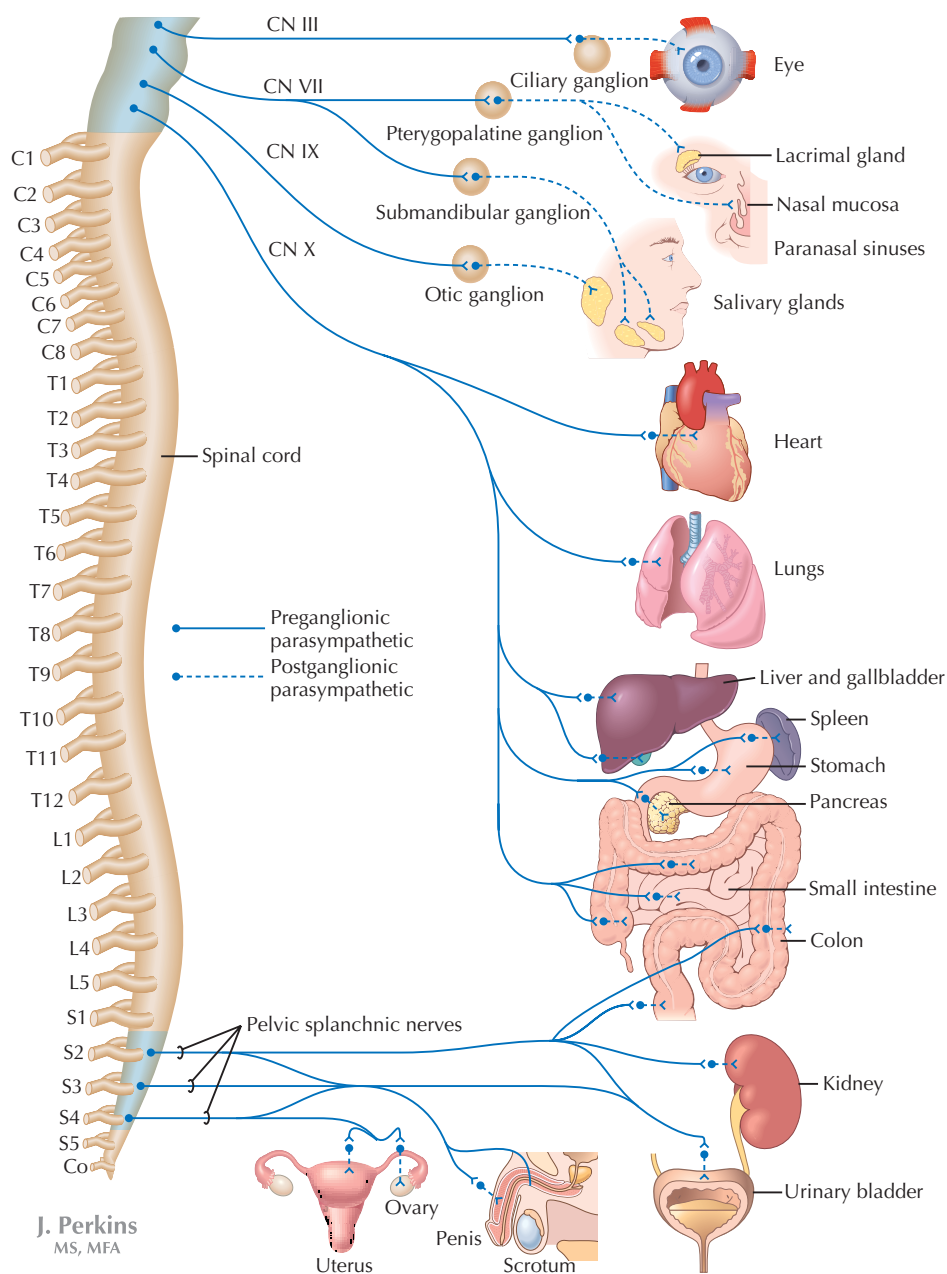


Sweat gland secretion is stimulated by activation of the sympathetic nervous system. Most of the postganglionic sympathetic neurons innervating these glands are atypical, releasing the neurotransmitter acetylcholine instead of norepinephrine. Acetylcholine acts on muscarinic receptors, inducing sweat secretion. However, in some specific areas, such as the palms of the hands, adrenergic nerves stimulate sweat glands through the release of norepinephrine, which acts at  $\alpha_1$  receptors to stimulate secretion.

## AUTONOMIC RECEPTORS

Autonomic receptors are coupled to specific G proteins within the cell membrane and produce their effects through various signal transduction systems in the effector cells. Acetylcholine receptors are classified as nicotinic or cholinergic, based on their pharmacology. Nicotinic receptors are activated by the drug **nicotine** and are blocked by **curare**, the active agent in South American dart poison. Muscarinic receptors, on the other hand, are activated by the mushroom toxin **muscarine**.





**Figure 7.3 Autonomic Nervous System: Parasympathetic Division** Preganglionic fibers of the PNS are associated with cranial nerves III, VII, IX, and X and also emerge from the sacral spinal cord at levels S2–S4. The PNS is involved in “vegetative” processes such as digestion as well as homeostatic functions. In general, the SNS and PNS work together in reciprocal fashion to regulate bodily function.

**Table 7.2** Actions of the Autonomic Nervous System

| Site of Action                 | PARASYMPATHETIC NERVOUS SYSTEM                                                                           |               | SYMPATHETIC NERVOUS SYSTEM                              |                     |
|--------------------------------|----------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------|---------------------|
|                                | Action                                                                                                   | Receptor type | Action                                                  | Receptor type       |
| Cardiac pacemaker              | Decreases heart rate                                                                                     | Muscarinic    | Increases heart rate                                    | $\beta_1$           |
| Cardiac muscle                 | Decreases contractility of atria; limited effects on ventricles                                          | Muscarinic    | Increased contractility                                 | $\beta_1$           |
| Cardiac AV node                | Decreases conduction velocity                                                                            | Muscarinic    | Increased conduction velocity                           | $\beta_1$           |
| Vascular smooth muscle         | Indirect vasodilation (genital organs and lower GI tract only) by nitric oxide released from endothelium | Muscarinic    | Constriction (predominant effect in most vascular beds) | $\alpha_1$          |
|                                |                                                                                                          |               | Vasodilation                                            | $\beta_2$           |
| Gastrointestinal smooth muscle | Increases motility                                                                                       | Muscarinic    | Reduces motility                                        | $\alpha_2, \beta_2$ |
|                                | Relaxes sphincters                                                                                       | Muscarinic    | Constricts sphincters                                   | $\alpha_1$          |
| Gastric parietal cells         | Acid secretion                                                                                           | Muscarinic    |                                                         |                     |
| Pancreas                       | Exocrine secretion                                                                                       | Muscarinic    |                                                         |                     |
| Lung, bronchial smooth muscle  | Constricts                                                                                               | Muscarinic    | Dilates                                                 | $\beta_2$           |
| Sweat glands                   |                                                                                                          |               | Secretion (generalized)                                 | Muscarinic*         |
|                                |                                                                                                          |               | Secretion (specific areas such as palms)                | $\alpha_1$          |
| Male reproductive system       | Erection                                                                                                 | Muscarinic    | Emission during orgasm                                  | $\alpha$            |
| Female reproductive system     | Vasocongestion, vaginal lubrication                                                                      | Muscarinic    | Orgasmic smooth muscle constriction                     | $\alpha$            |
| Pupil                          | Miosis (constriction)                                                                                    | Muscarinic    | Mydriasis (dilation)                                    | $\alpha_1$          |

AV, atrioventricular.

\*Atypical sympathetic cholinergic nerve fibers secrete acetylcholine.

and are blocked by the drug **atropine**, the deadly nightshade toxin. Nicotinic receptors are membrane channels that conduct  $\text{Na}^+$  and  $\text{K}^+$  when acetylcholine is bound. There are several subtypes of muscarinic receptors; the most common are coupled to G proteins that activate phospholipase C and produce elevation of  $\text{IP}_3$  and free intracellular  $\text{Ca}^{2+}$ .

Receptors for catecholamines are classified as  $\alpha$ -adrenergic or  $\beta$ -adrenergic on the basis of their responses to specific pharmacologic agents (drugs).  $\beta$ -receptors are blocked by the drug **propranolol**, and  $\alpha$ -receptors are blocked by **phentolamine**.

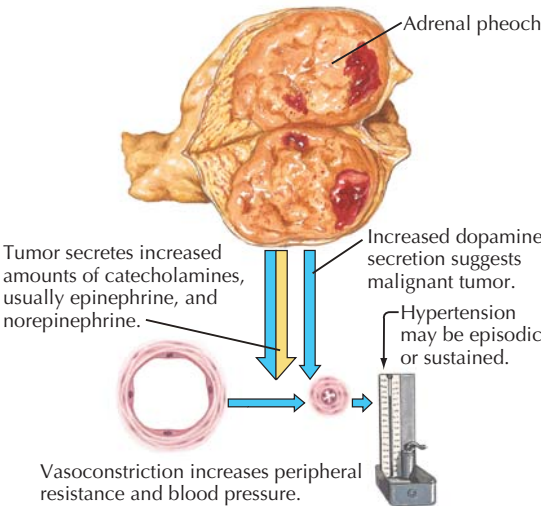
Adrenergic receptors may be further subclassified as  $\alpha_1$ ,  $\alpha_2$ ,  $\beta_1$ ,  $\beta_2$ , and  $\beta_3$  subtypes on the basis of effects of more specific drugs. At  $\alpha_1$ -receptors, binding of an agonist produces elevation of free intracellular  $\text{Ca}^{2+}$ ; binding of an agonist at  $\alpha_2$ -receptors produces inhibition of adenylyl cyclase and reduction of the levels of the second messenger cAMP. Activation of  $\beta$ -receptors results in activation of adenylyl cyclase and elevation of cAMP. How an organ or tissue responds to sympathetic stimulation is largely a function of the type of adrenergic receptors present in the tissue (see Table 7.2).

### CLINICAL CORRELATE

#### Pheochromocytoma

Tumors of the adrenal medulla that secrete epinephrine and norepinephrine are known as pheochromocytomas; catecholamine-secreting tumors in extraadrenal tissue may also occur. In either case, signs and symptoms are consistent with increased sym-

thetic activity. Catecholamines in plasma and 24-hour urine collections are elevated, and unlike catecholamines released by normal adrenal medulla, are not suppressed by clonidine administration (clonidine acts centrally to suppress sympathetic activity). Treatment is by surgical resection of the tumor. Pheochromocytoma is a rare disorder.



Adrenal pheochromocytoma

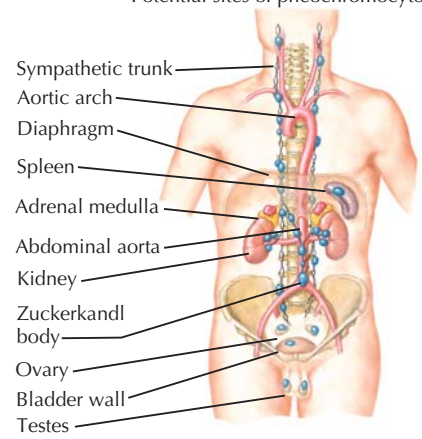
Tumor secretes increased amounts of catecholamines, usually epinephrine, and norepinephrine.

Increased dopamine secretion suggests malignant tumor.

Hypertension may be episodic or sustained.

Vasoconstriction increases peripheral resistance and blood pressure.

Potential sites of pheochromocytoma

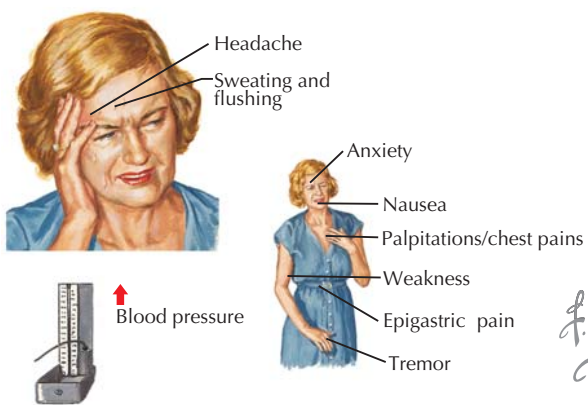


Sympathetic trunk  
Aortic arch  
Diaphragm  
Spleen  
Adrenal medulla  
Abdominal aorta  
Kidney  
Zuckerkindl body  
Ovary  
Bladder wall  
Testes

Pheochromocytoma is a chromaffin cell tumor secreting excessive catecholamines resulting in increased peripheral vascular resistance and hypertension.

Most pheochromocytomas are adrenal in origin, but can occur in various sites and may be associated with multiple endocrine neoplasia (MEN) syndromes. Most are sporadic, but some are hereditary.

#### Clinical features of pheochromocytoma



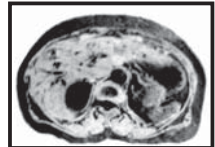
Headache  
Sweating and flushing  
Anxiety  
Nausea  
Palpitations/chest pains  
Weakness  
Epigastric pain  
Tremor

Blood pressure



Random urine sample      24-hour urine sample

Random urine assay for creatine and metanephrine or 24-hour urine assay of metanephrine and free catecholamines used in diagnosis.



CT scan or MRI may reveal presence of tumor.

Symptoms are secondary to excessive catecholamine secretion and are usually paroxysmal. More than 90% of patients with pheochromocytoma have headaches, palpitations, and sweating alone or in combination.

Catecholamines secreted by tumors of the adrenal medulla or other sites produce excess sympathetic-like activity.

# Review Questions

## CHAPTER 3: NERVE AND MUSCLE PHYSIOLOGY

1. The resting membrane potential of a cell is measured to be  $-70$  mV in a solution with ion concentrations resembling extracellular fluid. Which of the following manipulations would result in a hyperpolarization of the cell?

- A. Reduction in the membrane permeability to sodium ion
- B. Reduction in the membrane permeability to potassium ion
- C. Influx of calcium ion through the cell membrane
- D. Increase in the extracellular sodium ion concentration
- E. Increase in the extracellular potassium ion concentration

2. The rapid depolarization during phase 0 upstroke of the action potential in neurons is caused by opening of:

- A. ligand-gated  $\text{Ca}^{2+}$  channels.
- B. voltage-gated  $\text{Ca}^{2+}$  channels.
- C. voltage-gated  $\text{Na}^{+}$  channels.
- D. voltage-gated  $\text{K}^{+}$  channels.
- E. voltage-gated  $\text{Cl}^{-}$  channels.

3. In the myelinated regions of an axon,

- A. internal resistance is reduced.
- B. membrane resistance is reduced.
- C. the space constant is reduced.
- D. membrane capacitance is reduced.
- E. conduction velocity is reduced.

4. When depolarization of an axon reaches the synaptic bouton, release of neurotransmitter stored in presynaptic vesicles is most closely associated with:

- A. influx of  $\text{Ca}^{2+}$ .
- B. influx of  $\text{Na}^{+}$ .
- C. efflux of  $\text{Na}^{+}$ .
- D. influx of  $\text{K}^{+}$ .
- E. efflux of  $\text{K}^{+}$ .

5. Which of the following is primarily a characteristic of end-plate potentials at the neuromuscular junction, but not of neuronal action potentials?

- A. Depolarization is rapid
- B. Depolarization to a potential of  $+40$  mV
- C. Upstroke caused by opening of voltage-gated channels
- D. Charge is carried by a single, large channel for  $\text{Na}^{+}$  and  $\text{K}^{+}$
- E. Repolarization is associated with increased  $\text{K}^{+}$  conductance

## CHAPTER 4: ORGANIZATION AND GENERAL FUNCTIONS OF THE NERVOUS SYSTEM

6. The outermost tissue of the spinal cord is:

- A. white matter.
- B. gray matter.
- C. pia mater.
- D. the arachnoid membrane.
- E. dura mater.

7. Cerebrospinal fluid circulating through the lateral ventricles, third and fourth ventricles, and the subarachnoid space is produced by:

- A. the arachnoid membrane.
- B. the choroid plexus.
- C. the lymphatic system.
- D. arachnoid granulations.
- E. the pia mater.

8. Association pathways linking the two cerebral hemispheres are found in the:

- A. corpus callosum.
- B. anterior commissure.
- C. posterior commissure.
- D. hippocampal commissure.
- E. All of the above.

9. Centers involved in integration of cardiovascular and respiratory functions are mainly found in what region of the brainstem?

- A. Midbrain
- B. Pons
- C. Medulla oblongata
- D. Thalamus
- E. Hypothalamus

## CHAPTER 5: SENSORY PHYSIOLOGY

10. Which type of mechanoreceptor responds best to low frequency stimuli such as flutter?

- A. Meissner's corpuscles
- B. Pacinian corpuscles
- C. Merkel's disks
- D. Hair follicle receptors
- E. Ruffini's corpuscles



11. Light reception by photoreceptors results in reduction of permeability of what ion?

- A.  $K^+$
- B.  $Na^+$
- C.  $Cl^-$
- D.  $HCO_3^-$
- E.  $Ca^{2+}$

12. The endolymph of the scala media resembles which body fluid in its composition?

- A. Interstitial fluid
- B. Plasma
- C. CSF
- D. Lachrymal gland secretions
- E. Intracellular fluid

13. During angular acceleration of the head, neurotransmitter release by hair cells within the ampullary crest of the semicircular canal alters the firing rate of afferent nerve fiber axons through cranial nerve:

- A. IV.
- B. V.
- C. VI.
- D. VII.
- E. VIII.

## CHAPTER 6: THE SOMATIC MOTOR SYSTEM

14. The knee jerk reflex is an example of a:

- A. stretch reflex.
- B. Golgi tendon reflex.
- C. flexor withdrawal reflex.
- D. bisynaptic reflex.
- E. polysynaptic reflex.

15. The most important descending pathway(s) for control of fine motor movements originating in the cortex is (are) the:

- A. pontine reticulospinal tract.
- B. medullary reticulospinal tract.
- C. lateral and medial vestibulospinal tracts.
- D. tectospinal and rubrospinal tracts.
- E. corticospinal (pyramidal) tract.

16. The inhibitory efferent output of the cerebellar cortex emanates from which type of cell?

- A. Granule cells
- B. Golgi cells
- C. Purkinje cells
- D. Basket cells
- E. Stellate cells

17. Which of the following structures has as a primary function the regulation of the level of excitation in the motor cortex, resulting in smooth movement and maintenance of posture?

- A. Limbic system
- B. Basal ganglia
- C. The medulla oblongata
- D. Hypothalamus
- E. The pons

## CHAPTER 7: THE AUTONOMIC NERVOUS SYSTEM

18. Preganglionic nerve cell bodies in the sympathetic nervous system are located in the:

- A. cervical spinal cord.
- B. thoracolumbar spinal cord.
- C. sacral spinal cord.
- D. sympathetic chain ganglia.
- E. brainstem.

19. Release of acetylcholine and subsequent binding to muscarinic receptors is the major mode of neurotransmission at:

- A. parasympathetic ganglia.
- B. neuroeffector junctions of the SNS.
- C. sympathetic ganglia.
- D. neuroeffector junctions of the PNS.
- E. All of the above.

20.  $\beta_2$ -adrenergic receptors mediate which of the following effects of the sympathetic nervous system?

- A. Constriction of vascular smooth muscle
- B. Increased cardiac contractility
- C. Dilation of bronchial smooth muscle
- D. Dilation of the pupil (mydriasis)
- E. Sweat gland secretion

## Section 3

# CARDIOVASCULAR PHYSIOLOGY

*The main function of the cardiovascular system is transport of gases, nutrients, and wastes. It transports oxygen from the lungs to the rest of the body; carbon dioxide to the lungs from tissues throughout the body; nutrients between sites of absorption, utilization, and storage; and metabolic waste products from throughout the body to sites at which they are eliminated or recycled. The cardiovascular system also has important roles in thermoregulation, defense mechanisms, endocrine system function, and embryonic development. All of these roles are served by the continuous flow of blood, pumped by the heart, to capillary beds, where exchange between tissues and blood takes place. Nearly all living cells in our body are located no more than a few cell thicknesses from a capillary. In the following chapters, key concepts of cardiovascular physiology are presented, from basic electrophysiological and biophysical principles, to integrated mechanisms for control of cardiovascular system function.*

Chapter 8    **Overview of the Heart and Circulation**

Chapter 9    **Cardiac Electrophysiology**

Chapter 10   **Flow, Pressure, and Resistance**

Chapter 11   **The Cardiac Pump**

Chapter 12   **The Peripheral Circulation**

**Review Questions**

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# CHAPTER 8

## Overview of the Heart and Circulation

The main functions of the cardiovascular system are gas transport, nutrient delivery, and waste removal. These functions are dependent on circulation of blood, which requires structural integrity of the heart and vessels. In this chapter, the major structural features of the human heart and circulation are presented.

### GENERAL SCHEME OF THE CIRCULATION

The cardiovascular system consists of the heart, blood vessels, and blood. The overall circulation can be characterized as a series circuit (Fig. 8.1), in which:

- oxygenated blood is pumped by the left ventricle into the systemic arterial circulation,
- deoxygenated blood returns to the right atrium through the systemic veins,
- the right ventricle pumps this blood into the pulmonary circulation, and
- reoxygenated blood subsequently returns to the left atrium through the pulmonary veins.

Thus, all of the oxygenated blood returning to the left side of the heart from the lungs flows through the systemic circulation, and all of the deoxygenated blood returning to the right side of the heart is pumped to the pulmonary circulation for reoxygenation. Within these circulations, the regional circulations are parallel to each other. For example, skeletal muscle vascular beds lie in parallel to the renal vasculature. In both the systemic and pulmonary circulations, arteries are the *distributing* vessels, supplying blood to capillary beds of the body and lungs, respectively. The veins can be considered *collecting* vessels for return of capillary blood to the heart; they also serve a reservoir function, whereby generalized venous constriction produces an overall rise in blood pressure, for example during compensation for dehydration or blood loss. Of the total blood volume of 5 liters (L), more than 60% is found in the systemic veins (see Fig. 8.1).

### STRUCTURE OF THE HEART

The human heart consists of four chambers: the two muscular ventricles and the two atria. High-pressure ejection of blood

into the systemic circulation is accomplished by the action of the muscular left ventricle (Fig. 8.2). The interventricular septum functions as part of the left ventricle in this pumping action. The right ventricle ejects blood at lower pressures into the pulmonary circulation. The contraction of the left ventricle consists of constriction of the chamber and shortening of the heart from base to apex; the less forceful contraction of the right ventricle consists of a lateral to medial bellows-like motion, with some shortening from base to apex. In the left heart, backflow of blood into the atria is prevented by the **mitral valve** between the left ventricle and atrium. The right ventricle and atrium are separated by the **tricuspid valve**. The **chordae tendinae** are tendons connecting the **papillary muscles** to the mitral and tricuspid valves. The chordae tendinae and papillary muscles function to prevent eversion of the mitral and tricuspid valves into the atria during ventricular ejection, thereby preventing “regurgitation” of blood back into the atria. During ejection of blood, the aortic and pulmonary valves are open; they prevent backflow to the ventricles during cardiac filling. Valvular opening and closing are accomplished by passive responses of the valves to pressure gradients.



The four-chambered heart of mammals and birds is an evolutionary adaptation to prevent mixing of oxygenated blood returning to the heart from the lungs with deoxygenated blood returning from the systemic circulation. The efficiency of the series circulation created by this adaptation is consistent with the high metabolic rate and warm-blooded nature of these species. Some reptiles also have four-chambered hearts. Amphibians have a dual circulation with a three-chambered heart; the muscular ventricle is not separated into two pumps. In contrast to the dual circulation made possible by four chambers, fish have a single circulation and a two-chambered heart.

### CONDUCTION SYSTEM OF THE HEART

Effective ejection of blood from the heart requires coordinated contraction and relaxation of the chambers. At rest, this cycle occurs approximately 70 times per minute, and to accomplish this, the sinoatrial (SA) node acts as the pacemaker, initiating the depolarization of the heart.



Depolarization spreads from the SA node across the atria, depolarizing atrial muscle and initiating atrial contraction (Fig. 8.3). The atrioventricular (AV) node is the only pathway for spread of the depolarization to the ventricle. Electrical conduction through the AV node is slow, allowing atrial con-

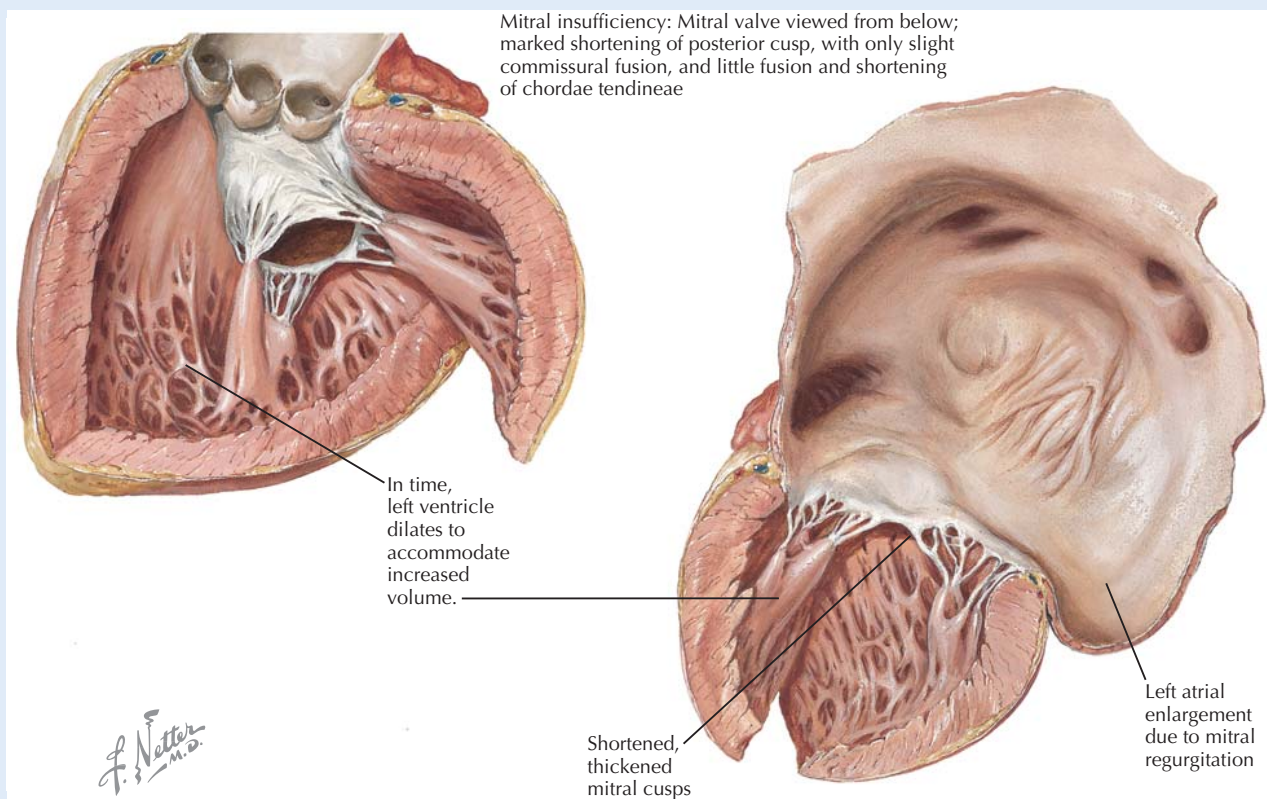
traction to accomplish the final filling of the ventricles. Next, depolarization reaches the bundle of His, and depolarization proceeds rapidly through the left and right bundle branches to the Purkinje fibers and ventricular muscle, producing ventricular contraction.

### CLINICAL CORRELATE Cardiac Structural Changes in Disease

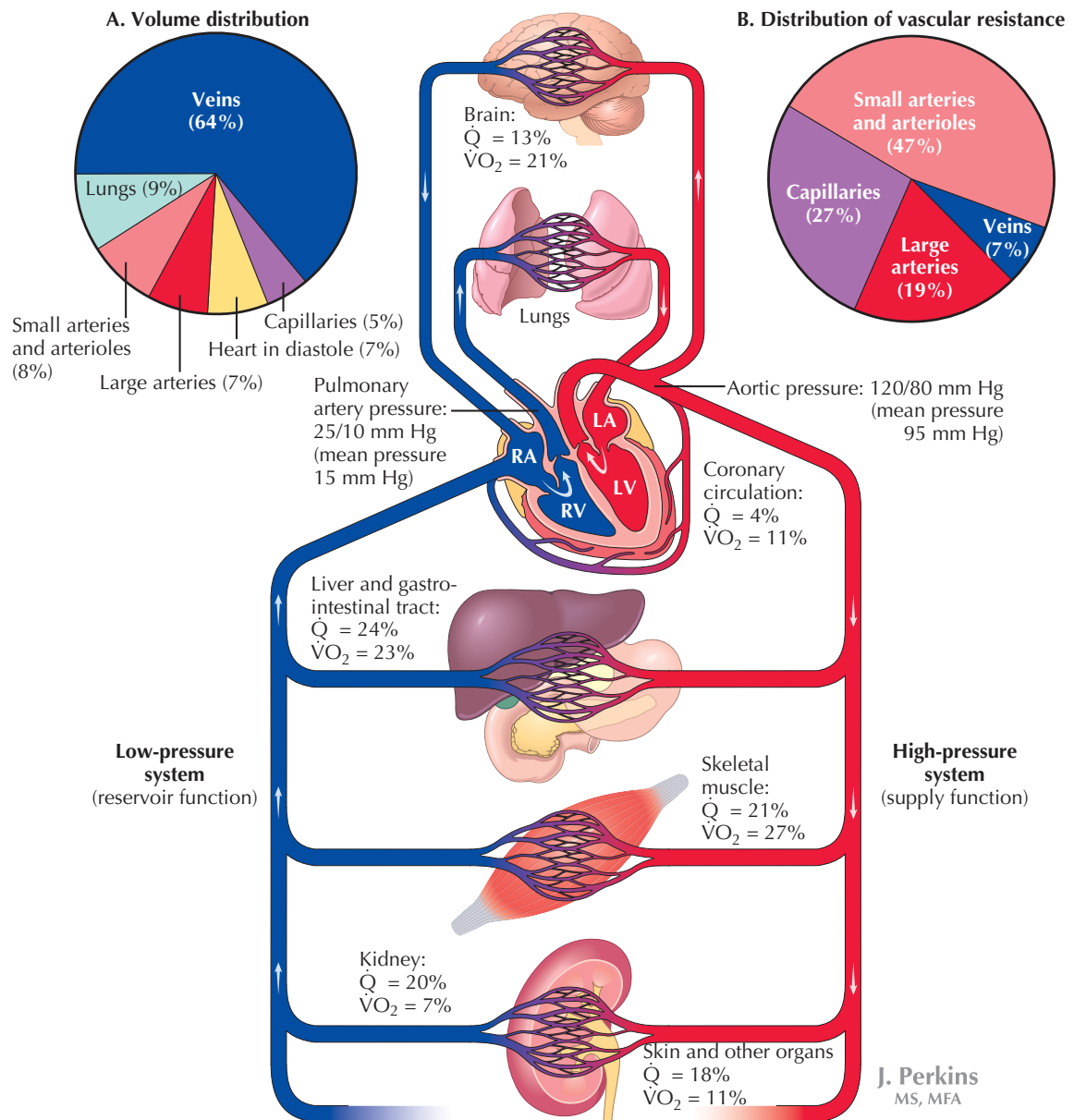
The normal structure of the heart may be altered in valvular disease, systemic or pulmonary hypertension, heart failure, and other diseases. Common pathologies affecting cardiac structure include the following:

- Pulmonary hypertension: Right ventricular hypertrophy may occur in response to the chronically increased work of the right ventricle.

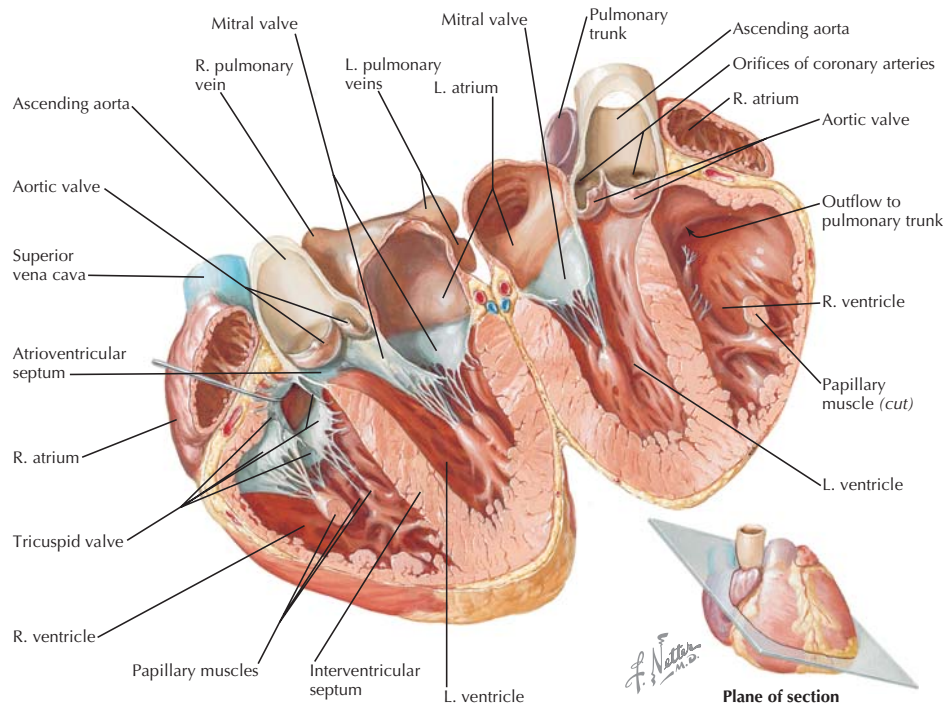
- Aortic stenosis (narrowing of the aortic valve): Left ventricular hypertrophy occurs in response to the greater work performed by the ventricle.
- Mitral incompetence: Left atrial dilation may develop as a result of the elevation of left atrial pressure and volume caused by mitral regurgitation (leakage of blood back into the atrium from the left ventricle).



**Cardiac Structural Changes in Mitral Regurgitation** Incompetence of the mitral valve results in regurgitation of blood into the left atrium during systole. Chronic mitral regurgitation results in left atrial enlargement and left ventricular dilation.

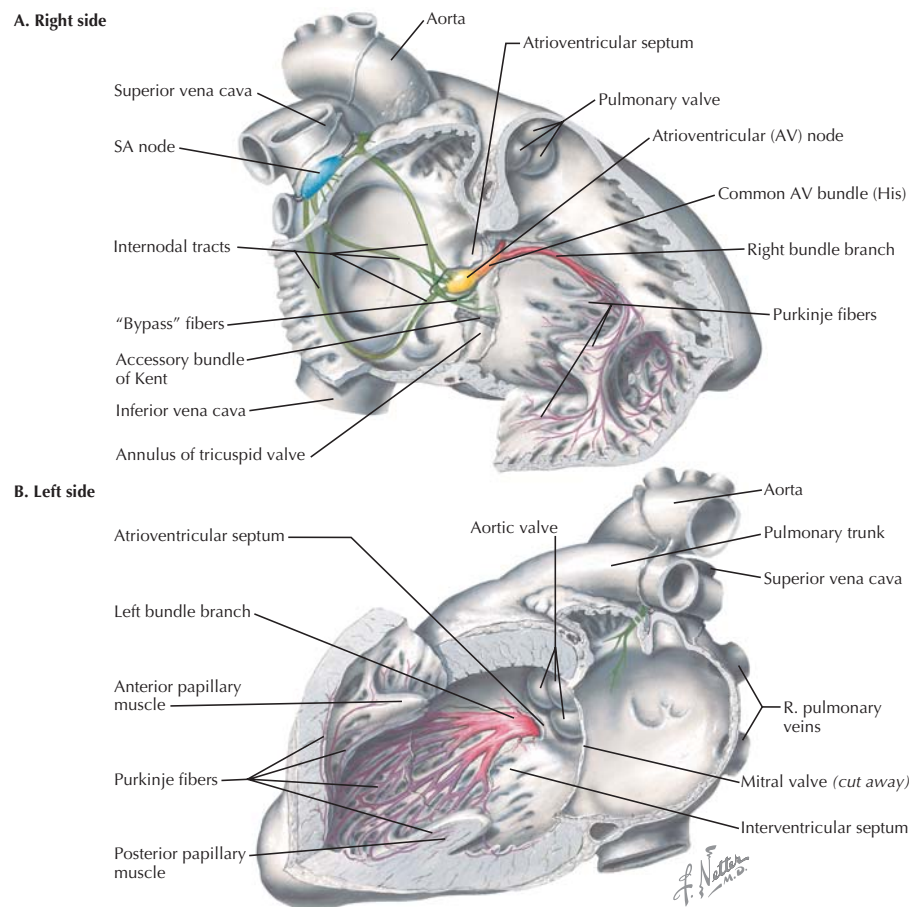


**Figure 8.1 General Scheme of the Circulation** The distribution of flow  $\dot{Q}$  to various organs as percentage of cardiac output and the utilization of oxygen  $\dot{V}O_2$  relative to total  $O_2$  consumption are illustrated. The pie charts show distribution of blood volume throughout the system (A) and the distribution of vascular resistance as a percentage of total resistance (B). The greatest volume of blood in the cardiovascular system is in the systemic veins. The greatest resistance to flow occurs in small arteries and arterioles. All values in the figure are those observed at rest; these are adjusted to meet changes in physiological requirements, for example during exercise.



**Figure 8.2 Structure of the Heart** The muscular left ventricle pumps blood into the high-pressure systemic circulation, and the less muscular right ventricle pumps blood into the lower pressure pulmonary circulation. The mitral and tricuspid valves prevent backflow of blood into the atria, while the aortic and pulmonic valves prevent regurgitation of blood from the aorta and pulmonary artery to the ventricles.

**Figure 8.3 Conduction System of the Heart** Contraction of cardiac muscle is initiated by electrical depolarization. The pacemaker for this electrical activity is in the sinoatrial (SA) node. Impulses travel from the SA node through internodal conducting pathways to the atrioventricular (AV) node, and then spread through the His-Purkinje system of the ventricle. The bypass fibers and Kent bundle are not normally part of the conducting system but may play a role in pathologic states.



# CHAPTER 9

## Cardiac Electrophysiology

Cells of the heart, like other excitable cells, have the ability to generate action potentials. These action potentials and their conduction throughout the heart are the basis of the rhythmic contraction and relaxation of the heart; the conduction of action potentials in the heart is recorded on the surface of the body as the electrocardiogram.

### CARDIAC ACTION POTENTIALS

In a normal heart, the pacemaker activity of the **sinoatrial (SA) node** produces a resting heart rate of approximately 70 beats per minute. The SA node is a group of specialized, non-contractile cardiomyocytes in the wall of the right atrium, adjacent to the opening of the superior vena cava. The resting membrane potential of SA node cells is initially around  $-60$  mV, but the cells undergo gradual, spontaneous depolarization, due to an inward current,  $i_b$ , mainly carried by  $\text{Na}^+$ ; an inward  $\text{Ca}^{2+}$  current,  $i_{Ca}$ ; and reduced activity of an outward  $\text{K}^+$  current. The *diastolic depolarization* resulting from the inward currents is responsible for the pacemaker activity of the SA node cells. Once threshold is reached, the upstroke of the action potential occurs, due to opening of T-type and L-type  $\text{Ca}^{2+}$  channels (Fig. 9.1). Repolarization is caused by increased conductance of  $\text{K}^+$  and closing of  $\text{Ca}^{2+}$  channels, completing the pacemaker cycle. Action potentials of **atrio-ventricular (AV) node cells**, found in the atrial septum at the junction of the atria and ventricles, are similar to those in the SA node, although the pacemaker activity of these cells is slower, overridden in normally functioning hearts by SA pacemaker activity.

In contrast to SA and AV nodal cells, depolarization of other cardiac cells is largely dependent on  $\text{Na}^+$  influx rather than  $\text{Ca}^{2+}$  influx. Action potentials of ventricular and atrial myocytes and cells of the His-Purkinje system are characterized by five distinct phases (see Fig. 9.1):

- **Phase 4 (the resting membrane potential):** The resting membrane potential of these cells is mainly a function of  $\text{K}^+$  efflux and is close to the Nernst potential for  $\text{K}^+$ . During phase 4, ion concentrations that were altered by the previous action potential are restored to resting levels by the  $\text{Na}^+/\text{K}^+$ -ATPase and a  $\text{Na}^+/\text{Ca}^{2+}$  exchanger, as well as an ATP-dependent  $\text{Ca}^{2+}$  pump.

- **Phase 0 (upstroke of the action potential):** Rapid depolarization occurs when the cells reach threshold and fast channels for  $\text{Na}^+$  open. This is accompanied by reduced conductance of the inwardly rectified  $\text{K}^+$  current ( $i_{K1}$ ). The rapid upstroke results in rapid spread of depolarization through most of the conducting system of the heart.
- **Phase 1 (rapid repolarization to the plateau):** This phase is caused by inactivation of  $\text{Na}^+$  channels and opening of voltage-sensitive  $\text{K}^+$  channels, producing a transient outward  $\text{K}^+$  current ( $i_{TO}$ ).
- **Phase 2 (the plateau):** The membrane remains depolarized during the plateau due to opening of voltage-sensitive, slow L-type  $\text{Ca}^{2+}$  channels and inward current of  $\text{Ca}^{2+}$ . Simultaneously, an outward  $\text{K}^+$  current occurs through a voltage-dependent  $\text{K}^+$  channel (delayed rectifier  $\text{K}^+$  channel).
- **Phase 3 (repolarization):** Gradual inactivation of the L-type  $\text{Ca}^{2+}$  channels leads eventually to activation of  $\text{K}^+$  channels, causing rapid repolarization, substantially due to an inwardly rectified  $\text{K}^+$  current ( $i_{K1}$ ).

The plateau of the action potential is functionally important, because it normally prevents premature depolarization of cardiac cells and associated arrhythmias. There is an **effective refractory period**, beginning with phase 1 and extending through the plateau and much of phase 3, during which another action potential cannot be generated. This is followed by the **relative refractory period**, which lasts until the resting membrane potential is fully restored; during this period, it is more difficult to elicit another action potential than in a resting cell.

During the conduction of the wave of depolarization of the heart, forward propagation occurs as a result of the local



Chronotropic, dromotropic, and inotropic effects are those that alter heart rate, conduction velocity, and myocardial contractility, respectively. A positive chronotropic agent is a drug that increases heart rate. Stimulation of the sympathetic nervous system increases the chronotropic, dromotropic, and inotropic state of the heart, whereas the parasympathetic nervous system mainly reduces the chronotropic and dromotropic state.



currents created during action potentials, and unidirectional depolarization is maintained due to the refractory periods of cells once they have depolarized.

## THE ELECTROCARDIOGRAM

An electrocardiogram (ECG or EKG) is a recording of changes in surface potential on the body, produced by the depolarization and repolarization of the heart. Electrocardiography may reveal the following:

- Abnormal cardiac rhythms and conduction.
- Presence, location, and extent of ischemia or infarction.
- Orientation of the heart in the thoracic cavity and size of chambers.
- Effects of abnormal electrolyte levels and some drugs.

These observations are facilitated by recording of the ECG with multiple lead configurations. A 12-lead ECG includes standard leads I, II, and III; six unipolar leads ( $V_1$ – $V_6$ ); and

three augmented limb leads (AVL, AVR, and AVF). A normal ECG consists of a:

- P wave, caused by atrial depolarization.
- QRS complex, representing ventricular depolarization.
- T wave, representing ventricular repolarization.

In the normal ECG, each P wave leads to a QRS complex and T wave. The heart is considered to be in “sinus rhythm,” because the rhythm is being regulated by the SA node. **Bradycardia** refers to a resting heart rate below 60 beats per minute; **tachycardia** is a resting heart rate above 100 beats per minute. Some causes of bradyarrhythmias in the presence of normal sinus rhythm include high vagal (parasympathetic) tone; drugs; and various metabolic, endocrine, and neurological disorders. Some causes of tachyarrhythmias with sinus rhythm include high sympathetic tone; drugs; low blood volume (dehydration); and metabolic, endocrine, and neurological disorders. The specific electrical events leading to the components of the ECG are detailed in Figures 9.2 to 9.4.

### CLINICAL CORRELATE Atrioventricular Block

The normal conducting pathway in the heart is:

SA → AV → Bundle of His → Bundle branches → Purkinje fibers → Ventricular muscle

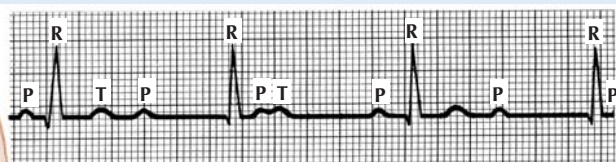
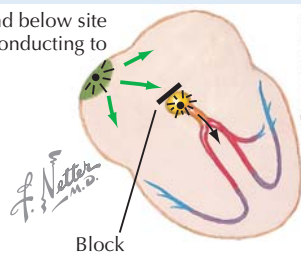
When conduction is disrupted in this pathway, arrhythmias result. Atrioventricular block (heart block) is a group of cardiac arrhythmias associated with altered conduction through the AV node:

- First-degree AV block is abnormally delayed conduction through the AV node (PR interval on the ECG exceeding 200 msec). Each P wave results in a QRS complex and thus ventricular contraction, as in normal sinus rhythm, but AV conduction is delayed. By itself, first-degree block does not produce bradycardia and is usually benign if not caused by underlying heart disease.
- Second-degree AV block is characterized by intermittent conduction through the AV node. As a result, some P waves on the ECG are followed by a QRS complex, whereas others are not.

Thus, the ventricles contract less frequently than the atria. Myocardial ischemia or infarction may produce second-degree block.

- Third-degree AV block (complete block) occurs when the wave of depolarization is not conducted through the AV node (from atria to ventricles). On the ECG, both P-P intervals and R-R intervals are regular, but P waves are dissociated from the QRS complexes. When atrial pacemaker activity fails to be conducted to the ventricle, an escape pacemaker may arise in the AV node below the site of block or in the bundle of His, producing a ventricular rate of 40 to 55 beats per minute (as determined by the R-R interval) that is only partially responsive to sympathetic stimulation. When the conduction block occurs in the bundle of His, the ventricular escape rhythm will produce an inadequate heart rate of 20 to 40 beats per minute; cardiac output (flow rate from one ventricle) may be insufficient even at rest, and activity will be greatly restricted due to inability to adequately adjust output. The usual therapy is implantation of a cardiac pacemaker.

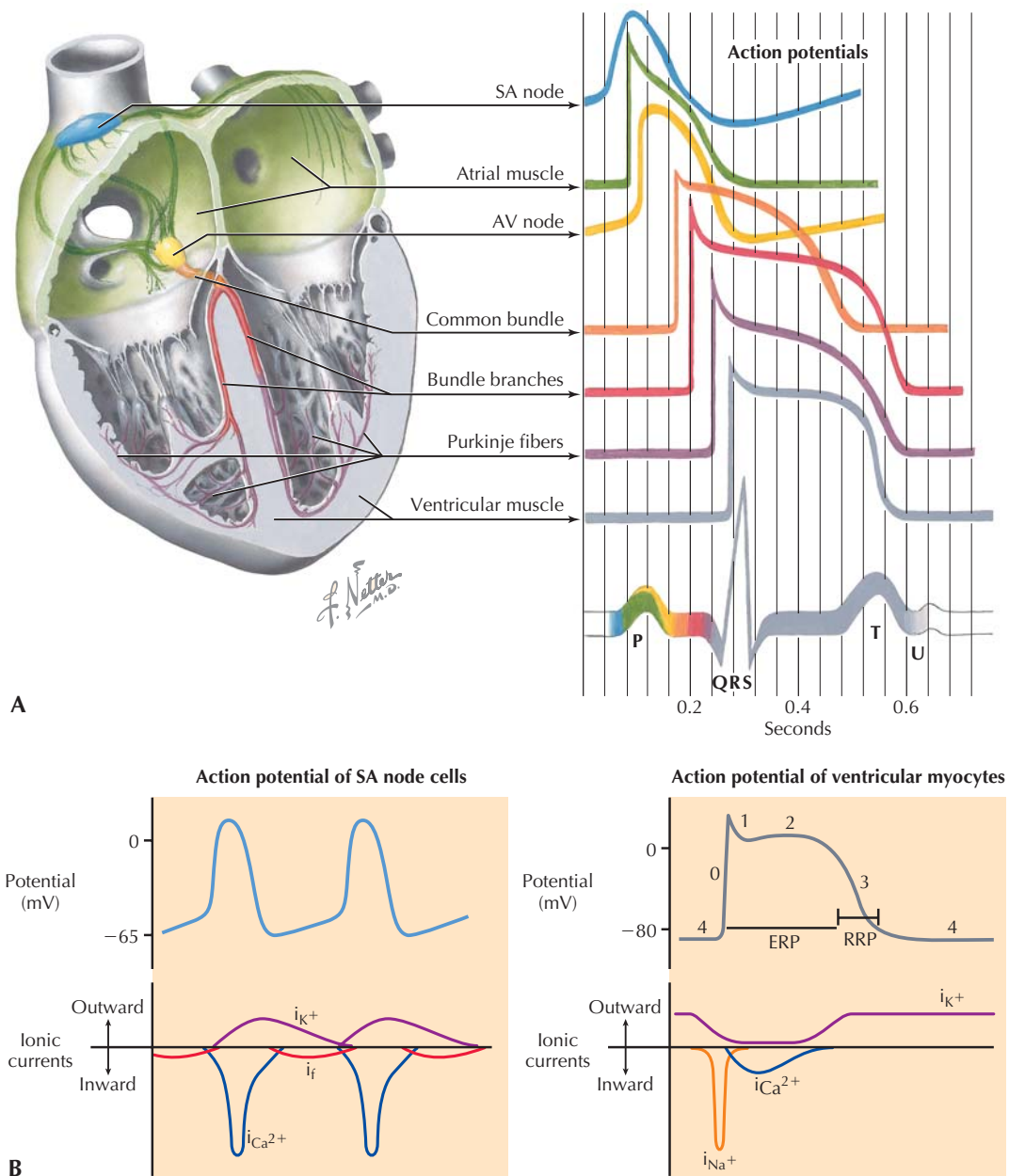
Impulses originate at SA node (P waves) and below site of block in AV node (junctional rhythm), conducting to ventricles.



Atria and ventricles depolarize independently. QRS complexes less frequent; regular at 40 to 55 beats/min but normal in shape.

### Third-Degree Heart Block





**Figure 9.1 Electrical Activity of the Heart** Action potentials vary as the wave of depolarization proceeds through the conducting pathway of the heart, beginning at the SA node and ending in the ventricular muscle. The ECG records changes in surface potential on the body associated with depolarization and repolarization of the heart. P waves, QRS complexes, and T waves constitute the normal ECG. In 50% to 70% of individuals, a U wave will also be recorded. Its origin is not well defined (A). Differences in morphology of the action potentials are based on differences in specific ion currents (B). In AV nodal cells, note the spontaneous depolarization from the resting membrane potential that results in the action potential. In ventricular myocytes (and also the His-Purkinje system), note the sharp phase 0 upstroke that produces rapid conduction of the wave of depolarization. The phase 2 plateau is important in preventing premature depolarization. The effective refractory period (ERP) is the period during which another action potential cannot be elicited; during the relative refractory period (RRP), it is more difficult to elicit an action potential than during phase 4.